Robust Precoder Design for Massive MIMO High-Speed Railway Communications With Matrix Manifold Optimization

Rui Sun, *Graduate Student Member, IEEE*, Chen Sun, *Member, IEEE*, Ding Shi, *Member, IEEE*, An-An Lu, *Member, IEEE*, Xiqi Gao, *Fellow, IEEE*, and Xiang-Gen Xia, *Fellow, IEEE*

***Abstract*— In high-speed railway (HSR) communications, the channel suffers from severe Doppler and channel aging effects caused by the high mobility, making the channel outdated quickly. To address this issue, we investigate the robust precoder design against channel aging and prediction inaccuracy in massive multiple-input multiple-output (MIMO) systems with matrix manifold optimization. First of all, we introduce the concept of the quadruple beams (QBs), and establish a QB based channel model with sampled quadruple steering vectors. Then, the upcoming space domain channel of interest can achieve a higher accuracy by channel prediction with the estimated QB domain channel. To further improve the performance while save the pilot overhead, we predict the forthcoming QB domain channel and integrate the prediction inaccuracy within the *a posterior* QB domain statistical channel model. Then, we consider the robust precoder design aiming to maximize the upper bound of the ergodic weighted sum-rate (WSR) on the Riemannian submanifold formed by the precoders satisfying the total power constraint (TPC). Riemannian ingredients are derived for matrix manifold optimization, with which the Riemannian conjugate gradient (RCG) method is proposed to solve the unconstrained problem on the manifold. The RCG method mainly involves the matrix multiplication and avoids the need of matrix inversion of the transmit antenna dimension. The simulation results demonstrate the effectiveness of the proposed channel model and the superiority of the RCG method for robust precoder design against channel aging and prediction inaccuracy.**

***Index Terms*— High-speed railway, massive MIMO, manifold optimization, robust precoding, Riemannian submanifold, weighted sum-rate.**

Received 23 August 2024; revised 24 December 2024; accepted 9 February 2025. Date of publication 21 February 2025; date of current version 12 May 2025. This work was supported in part by Jiangsu Province Major Science and Technology Project under Grant BG2024005; in part by the National Natural Science Foundation of China under Grant 62271145, Grant 62394294, and Grant 62371125; in part by Jiangsu Province Basic Research Project under Grant BK20192002; in part by the Fundamental Research Funds for the Central Universities under Grant 2242023K5003 and Grant 2242022k60007; in part by the Key Research and Development Plan of Jiangsu Province under Grant BE2022067 and Grant BE2022067-2; and in part by the Huawei Cooperation Project. An earlier version of this paper was presented in part at the 2024 IEEE 24th International Conference on Communication Technology (ICCT). The associate editor coordinating the review of this article and approving it for publication was X. Cao. *(Corresponding authors: Xiqi Gao; Ding Shi.)*

Rui Sun, Chen Sun, An-An Lu, and Xiqi Gao are with the National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China, and also with the Purple Mountain Laboratories, Nanjing 211111, China (e-mail: ruisun@seu.edu.cn; sunchen@seu.edu.cn; aalu@seu.edu.cn; xqgao@seu.edu.cn).

Ding Shi was with the National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China. He is now with the Purple Mountain Laboratories, Nanjing 211111, China (e-mail: shiding@pmlabs.com.cn).

Xiang-Gen Xia is with the Department of Electrical and Computer Engineering, University of Delaware, Newark, DE 19716 USA (e-mail: xxia@ee.udel.edu).

Digital Object Identifier 10.1109/TWC.2025.3541644

I. Introduction

With the rapid deployment of the high-speed railway (HSR) throughout the world, communication systems tailored for HSR have gained great attention from both academia and industry [1], [2], [3]. Meanwhile, fueled by the continuous evolution of wireless technologies, the pursuit for increased data rates and enhanced quality services also manifests an exponential growth trend in HSR communications [4], [5], [6]. Massive multi-input multi-output (MIMO) is one of the most important technologies to deliver these urgent demands for the HSR communications [7], [8]. With a large number of antennas equipped at the base station (BS) side, massive MIMO can serve numerous users simultaneously on the same time-frequency resource [9], thereby reaping multiplexing gains and providing considerable enhancements in spectral and energy efficiencies [10], [11].

However, accommodating a number of users simultaneously inevitably causes severe inter-user interference [12], [13]. To deal with this challenge, precoding is used to suppress the interference and improve the received signal strength, and has been widely investigated in the HSR communications [14], [15], [16]. To be specific, hybrid precoding is adaptively invoked for millimeter wave communication to deal with random blockages in [14]. In [15], a novel hybrid precoding scheme is developed based on a non-uniform beam design to minimize the rate gap between different locations by selecting the appropriate predefined beams. To realize a better sum rate and steady communication, the authors in [16] propose a reconfigurable intelligent surface (RIS)-aided hybrid precoding for anti-blocking HSR network.

In practice, the downlink precoder design highly depends on the available channel state information (CSI) at the BS side [17], [18]. Most existing works assume the instantaneous CSI is perfectly known at the BS side [14], [15], [16], [19], [20], which, however, is not realistic in practice, especially in HSR communications [21], [22], [23]. Generally speaking,

1536-1276 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

the channel is scheduled to be estimated periodically at the pilot segments via the uplink sounding process [24], and then applied to the segments within the coherent time to ensure the accuracy while avoid excessive pilot overhead [25]. Nevertheless, the channel estimated from the pilot segment becomes outdated quickly due to the severe Doppler effect induced by the high mobility [26], [27], leading to the so-called channel aging effect [23]. To address this issue, channel prediction is identified as an enabling technique to obtain a more precise CSI without increasing the pilot overhead by exploiting the temporal correlation between channels [28], [29]. To better model this temporal correlation caused by the Doppler effect, it is proposed to introduce the concept of triple beams (TBs) in the space-frequency-time (SFT) domain and build a TB based channel model with sampled triple steering vectors [30] within the orthogonal frequency division multiplexing (OFDM) transmission framework. By further considering the angle of arrival (AoA), the TB based channel model can be extended to the quadruple-beam (QB) based channel model, which is more realistic in light of the burgeoning deployment of multi-antenna configurations in user terminals (UTs) [31]. Compared to the space domain channel, the QB domain channel exhibits relatively slower variation and greater sparsity in HSR communications. Therefore, the time varying space domain channel can be predicted via the relationship between the space domain channel and the QB domain channel, providing a more accurate CSI at the segment for data transmission.

In most cases, the predicted channel has demonstrated much better accuracy compared to the channel estimated from the pilot segment [30], but there still remains room for improvement by considering the prediction inaccuracy caused by the limited resolution in the Doppler domain. This inaccuracy could become more evident as the temporal intervals between the data symbol to be predicted and the pilot symbol for estimation extends [30]. It is proposed that the channel inaccuracy caused by the prediction error can be modeled as the channel uncertainty, which can be further constructed as a complex Gaussian random matrix with independent and identically distributed (i.i.d.), zero mean and different variance entries [18], [24], [32]. In addition, the sliding window technique is widely utilized in practice [33], and the QB domain channel needs to be estimated every time a new pilot segment arrives, indicating that the QB domain channels actually form a time series with strong temporal correlation. Therefore, inspired by the *a posterior* angle beam based statistical channel model proposed in [25], we take the prediction inaccuracy into consideration and establish *a posterior* QB based statistical channel model to describe the channel uncertainty more precisely via the popular Gauss-Markov process [12], [18], [24], [25], [32], with which the spatial channel between the pilot segments can be better predicted.

Matrix manifold optimization has been widely investigated thanks to its ability of transforming a constrained problem in Euclidean space into an unconstrained one on a manifold [34], [35], [36], [37]. It has been shown in our previous work that the Riemannian conjugate gradient (RCG) method for precoder design with perfect CSI exhibits a fast convergence speed and superior performance compared to the existing methods [38]. More importantly, the RCG method can avoid the matrix inversion over the transmit antenna dimension, which is of increasingly benefits in the future wireless networks where more antennas will be equipped at the BS side [39]. Therefore, we are motivated to investigate the RCG method for robust precoder design for HSR communications with the *a posterior* QB domain channel.

In this paper, we investigate the robust precoder design against channel aging and prediction error for massive MIMO HSR communications with matrix manifold optimization. The main contributions are summarized as follows.

- We establish a QB based channel model from the physical principle of HSR channels with sampled quadruple steering vectors in the SFT domain, each of which corresponds to a physical QB consisting of the AoA beam, the angle of departure (AoD) beam, the frequency-beam, and the temporal-beam, pointing towards the sampled directional cosine of AoA, directional cosine of AoD, delay, and Doppler frequency, respectively. Then, the space domain channel at any subcarrier and symbol of interest can be predicted against channel aging by virtue of the dependancy between the QB domain channel and the space domain channel.
- To further improve the performance of the predicted channel, we treat the QB domain channels periodically estimated from the pilot segments as a temporally correlated time series. With the widely used Gauss-Markov process, the *a posterior* QB based statistical channel model that incorporates the channel uncertainty caused by the prediction inaccuracy is further proposed for robust precoding.
- To avoid complicated expectation operators involved in the ergodic rate, we maximize the upper bound of the ergodic weighted sum-rate (WSR) under total power constraint (TPC) for robust precoder design. Then, the constrained problem is reformulated as an unconstrained one on the Riemannian submanifold formed by the precoder set satisfying TPC, and the Riemannian ingredients are derived for optimization on the manifold. With the derived Riemannian ingredients, the RCG method is proposed to solve the unconstrained problem.
- The complexity analyses show that our proposed method has a much lower computational complexity than the existing methods, especially with a large number of antennas at the BS side. The simulation results demonstrate the effectiveness of our proposed channel model and validate the superiority of the RCG method for the robust precoder design against channel aging and prediction inaccuracy.

The rest of this paper is organized as follows. In Section II, we first clarify the system configuration and establish the QB based channel model for HSR communications in the massive MIMO system. Then, the channel prediction and the *a posterior* QB based channel model is further derived for obtaining a more accurate CSI for precoding. The problem for robust precoder design is formulated on the Riemannian submanifold, and the Riemannian ingredients of the

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

Riemannian submanifold needed in matrix manifold optimization are derived in Section III. Section IV presents the RCG design method and the complexity analysis. Simulation results are provided in Section V to validate the superiority of our proposed QB based channel model and the RCG method for robust precoder design. The conclusion is drawn in Section VI.

Notations: Boldface lowercase and uppercase letters represent the column vectors and matrices, respectively. We write conjugate transpose of matrix $\mathbf{A}$ as $\mathbf{A}^H$ while $\mathrm{tr}(\mathbf{A})$ and $\det(\mathbf{A})$ denote the matrix trace and determinant of $\mathbf{A}$, respectively. $\Re\{\mathbf{A}\}$ means the real part of $\mathbf{A}$. $\mathrm{vec}(\mathbf{A})$ yields a vector $\mathbf{a} \in \mathbb{C}^{MN\times 1}$, which is obtained by stacking the columns of $\mathbf{A}$. $\mathrm{unvec}(\mathbf{a}, (M, N))$ yields a matrix $\mathbf{A} \in \mathbb{C}^{M\times N}$. Let the mathematical expectation be $\mathbb{E}\{\cdot\}$. $\mathbf{I}_M$ denotes the $M \times M$ identity matrix, whose subscript may be omitted for brevity. $\mathbf{0}$ represents the vector or matrix whose elements are all zero. $\mathrm{diag}(\mathbf{a})$ represents the diagonal matrix with $\mathbf{a}$ along its main diagonal and $\mathrm{diag}(\mathbf{A})$ denotes the column vector of the main diagonal of $\mathbf{A}$. Similarly, $\mathbf{D} = \mathrm{blkdiag}\{\mathbf{A}_1, \cdots, \mathbf{A}_K\}$ denotes the block diagonal matrix with $\mathbf{A}_1, \cdots, \mathbf{A}_K$ on the diagonal and $[\mathbf{D}]_i$ denotes the i-th matrix on the diagonal, i.e., $\mathbf{A}_i$. $(\mathbf{A})_{i,j}$ denotes the (i, j)-th element of $\mathbf{A}$. $|a|$ denotes the modulus of a.

II. Channel Modeling and Channel Prediction

A. System Configuration

We consider a single-cell massive MIMO-OFDM system in the HSR scenario. Specifically, we focus on the most common viaduct scenario in HSR communications, which makes up 40% for a typical railway line and is dominated by the line of sight (LoS) component [40].

The BS serves U UTs in the HSR that moves at the speed of v km/h in the viaduct scenario. $\mathcal{U} = \{1, 2, \cdots, U\}$ represents the set of all the UTs. There are M_t and M_r uniform linear antennas (ULA) equipped at the BS side and each UT side, respectively. As shown in Fig. 1, there are totally P_u paths from the BS to UT u, including one LoS path and $(P_u - 1)$ non-line of sight (NLoS) paths. All the NLoS paths of UT u constitute a set $\mathcal{S}_u^{\mathrm{NLoS}}$. The p-th path is composed of $Q_{p,u}$ subpaths, which are termed as the subpath cluster of path p. Particularly, $Q_{\mathrm{LoS},u} = 1$ for the LoS path.

The massive MIMO-OFDM system for HSR communications operates in time-division duplex (TDD) mode with frame structure shown in Fig. 2. Each subframe consists of N_b time slots and each slot is made up of N_s OFDM symbols. Let $N = N_b N_s$ denote the total number of the OFDM symbols in a subframe. Each OFDM symbol has N_c subcarriers, where N_v valid sub-carriers are used for data transmission with the index set of $\mathcal{S}_v = (m_0, m_1, \cdots, m_{N_v-1})$. Let f_c, $\lambda_c = \frac{c}{f_c}$ and Δf represent the center frequency, wave length and the subcarrier spacing, respectively. The system sampling interval is $T_s = \frac{1}{N_c \Delta f}$ and the length of cyclic prefix (CP) is N_g. $T_g = N_g T_s$ and $T_{\mathrm{sym}} = (N_c + N_g) T_s$ denote the time duration of CP and OFDM symbol including CP, respectively. Generally, it is assumed that the CP duration T_g is larger than the maximum channel delay of each UT.

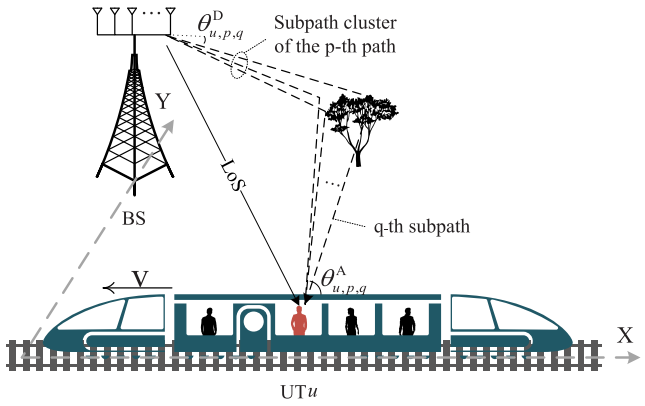


Fig. 1. The massive MIMO system in HSR.

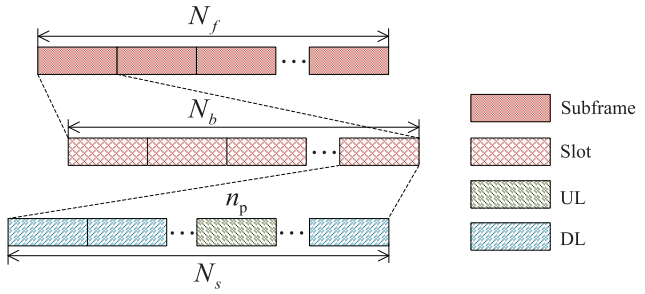


Fig. 2. The frame structure.

We assume the channel parameters, including the Doppler shift, time delay, AoD and AoA, exhibit minimal variation in each subframe. In addition, we assume that the CSI keeps constant within an OFDM symbol time duration T_{sym} and varies symbol by symbol due to the severe Doppler effect. For simplicity, it is assumed that there only exist the uplink training phase and the downlink transmission phase. In each slot, the uplink sounding reference signals are sent once at the n_{p}-th OFDM symbol, and the rest symbols are used for downlink (DL) transmission. Assuming that the CSI is perfectly known at the user side via the DL training phase, which is included in the DL data transmission and omitted in the slot structure for simplicity.

B. QB Based Channel Model

Let $\mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right)$ denote the user antenna response of the m_r-th antenna of UT u along the q-th subpath of the p-th path corresponding to the AoA $\theta^A_{u,p,q}$, where $\Omega^A_{u,p,q} = \cos\left(\theta^A_{u,p,q}\right)$. Let $\mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right)$ signify the BS antenna response of the m_t-th antenna of UT u along the q-th subpath of p-th path corresponding to the AoD with $\Omega^D_{u,p,q} = \cos\left(\theta^D_{u,p,q}\right)$. $\mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right)$ and $\mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right)$ can be written as

$$\mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right) = e^{-\jmath 2\pi m_r \Delta_r \Omega^A_{u,p,q}}, \tag{1a}$$

$$\mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right) = e^{-\jmath 2\pi m_t \Delta_t \Omega^D_{u,p,q}}, \tag{1b}$$

respectively, where Δ_r and Δ_t are the normalized receive antenna separation and transmit antenna separation, respectively. Let $\alpha_{u,p,q}$, $\tau_{u,p,q}$ and $\nu_{u,p,q}$ represent the propagation loss, propagation delay and the Doppler shift of the

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

u-th UT along the q-th subpath of the p-th path, respectively. The Doppler shift primarily arises from the mobility of the HSR, while the mobility of users within the carriages can be neglected. We assume that all the subpaths of a dominant path share the same time delay and Doppler shift. Then, we have $\tau_{u,p} = \tau_{u,p,q}$ and $\nu_{u,p} = \nu_{u,p,q}$.

Recall that the viaduct scenario is typically dominated by the LoS path. Due to the regular movements of high-speed train, the Doppler shift of the LoS path can be efficiently compensated to mitigate the inter-carrier interference (ICI) [41], [42]. Define $\gamma_{u,p,q} = \alpha_{u,p,q} * e^{-\jmath 2\pi f_c \tau_{u,p}}$ as the complex-valued gain. The channel impulse response between the m_t-th transmit antenna and the m_r-th receive antenna of UT u after Doppler compensation is then given by

$$\bar{h}_{u,m_r,m_t}(t,\tau) = \sum_{p=1}^{P_u}\sum_{q=1}^{Q_{p,u}} \gamma_{u,p,q} \mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right) \mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right) \times e^{\jmath 2\pi \nu^{\mathrm{cps}}_{u,p} t} \delta\left(\tau - \tau_{u,p}\right), \tag{2}$$

where $\nu^{\mathrm{cps}}_{u,p} = \nu_{u,p} - \nu^{\mathrm{LoS}}_u$ is the Doppler frequency after compensation with ν^{LoS}_u being the Doppler shift of the LoS path of UT u. After the OFDM operation, the channel frequency response at the m-th subcarrier of the n-th OFDM symbol between the m_t-th transmit antenna and the m_r-th receive antenna of UT u can be written as

$$\begin{aligned} h^{\mathrm{SFT}}_{u,m_r,m_t,m,n} &= \int \bar{h}_{u,m_t,m_r}(nT_{\mathrm{sym}},\tau)\, e^{-\jmath 2\pi m \Delta f \tau} d\tau \\ &= \sum_{p=1}^{P_u}\sum_{q=1}^{Q_{p,u}} \gamma_{u,p,q} \mathrm{u}_{m_r}\left(\Omega^A_{u,p,q}\right) \mathrm{v}_{m_t}\left(\Omega^D_{u,p,q}\right) \\ &\quad \times e^{-\jmath 2\pi m \Delta f \tau_{u,p}} e^{\jmath 2\pi \nu^{\mathrm{cps}}_{u,p} n T_{\mathrm{sym}}}. \end{aligned} \tag{3}$$

Based on (3), we further consider the whole space-frequency-time (SFT) domain channel over N OFDM symbols between the BS and UT u, which is termed as the SFT domain channel vector $\mathbf{h}^{\mathrm{SFT}}_u$, with element $h^{\mathrm{SFT}}_{u,m_r,m_t,m,n}$ of index $(nN_vM_tM_r + (m-m_0)M_tM_r + m_tM_r + m_r)$. Let

$$\begin{aligned} \mathbf{u}\left(\Omega^A\right) &= [1, e^{-\jmath 2\pi \Delta_r \Omega^A}, \cdots, e^{-\jmath 2\pi (M_r-1)\Delta_r \Omega^A}], \\ \mathbf{v}\left(\Omega^D\right) &= [1, e^{-\jmath 2\pi \Delta_t \Omega^D}, \cdots, e^{-\jmath 2\pi (M_t-1)\Delta_t \Omega^D}], \\ \boldsymbol{f}(\tau) &= \left[e^{-\jmath 2\pi m_0 \Delta f \tau}, e^{-\jmath 2\pi m_1 \Delta f \tau}, \cdots, e^{-\jmath 2\pi m_{N_v-1} \Delta f \tau}\right], \\ \mathbf{d}(\nu) &= \left[1, e^{\jmath 2\pi \nu T_{\mathrm{sym}}}, \cdots, e^{\jmath 2\pi \nu (N-1) T_{\mathrm{sym}}}\right] \end{aligned} \tag{4}$$

denote the steering vectors in the space domain of the receive antenna, the space domain of the transmit antenna, the frequency domain and the time domain, respectively. Note that each path of the channel is characterized by the tuple $\left(\Omega^A, \Omega^D, \tau, \nu\right)$. Denote

$$\mathbf{s}\left(\Omega^A, \Omega^D, \tau, \nu\right) = \mathbf{d}(\nu) \otimes \left\{\boldsymbol{f}(\tau) \otimes \left[\mathbf{v}\left(\Omega^D\right) \otimes \mathbf{u}\left(\Omega^A\right)\right]\right\} \tag{5}$$

as the SFT domain steering vector. In this way, the SFT domain channel can be expressed as

$$\mathbf{h}^{\mathrm{SFT}}_u = \sum_{p=1}^{P_u}\sum_{q=1}^{Q_{p,u}} \gamma_{u,p,q} \mathbf{s}\left(\Omega^A_{u,p,q}, \Omega^D_{u,p,q}, \tau_{u,p}, \nu^{\mathrm{cps}}_{u,p}\right). \tag{6}$$

The channel parameters $\Omega^A_{u,p,q}$, $\Omega^D_{u,p,q}$, $\tau_{u,p}$ and $\nu^{\mathrm{cps}}_{u,p}$ are limited in the sets $\mathcal{S}_{\mathrm{r}} = \left\{\Omega^A \mid \Omega^A \in [-1,1)\right\}$, $\mathcal{S}_{\mathrm{t}} = \left\{\Omega^D \mid \Omega^D \in [-1,1)\right\}$, $\mathcal{S}_{\mathrm{de}} = \left\{\tau \mid \tau \in [0, \tau_{\max})\right\}$ and $\mathcal{S}^u_{\mathrm{do}} = \left\{\nu \mid \nu \in \left[-\nu_{\max} - \nu^{\mathrm{LoS}}_u, \nu_{\max} - \nu^{\mathrm{LoS}}_u\right)\right\}$, respectively, where $\tau_{\max} = \frac{N_g}{N_c \Delta f}$ is the maximum delay spread and $\nu_{\max} = \frac{\mathrm{v}}{\lambda_c} = \frac{N_d}{2NT_{\mathrm{sym}}}$ is the maximum Doppler shift [30]. We uniformly divide these sets into multiple disjoint subsets as

$$\mathcal{S}_r = \bigcup_{n_r=0}^{N_r-1} \mathcal{S}^r_{n_r}, \mathcal{S}^r_{n_r} = \left[\frac{2n_r - N_r}{N_r}, \frac{2(n_r+1) - N_r}{N_r}\right), \tag{7a}$$

$$\mathcal{S}_t = \bigcup_{n_t=0}^{N_x-1} \mathcal{S}^t_{n_t}, \mathcal{S}^t_{n_t} = \left[\frac{2n_t - N_t}{N_t}, \frac{2(n_t+1) - N_t}{N_t}\right), \tag{7b}$$

$$\mathcal{S}_{\mathrm{de}} = \bigcup_{n_{\mathrm{de}}=0}^{N_{\mathrm{de}}-1} \mathcal{S}^{\mathrm{de}}_{n_{\mathrm{de}}}, \mathcal{S}^{\mathrm{de}}_{n_{\mathrm{de}}} = \left[\frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}} N_v \Delta f}, \frac{N_\tau (n_{\mathrm{de}}+1)}{N_{\mathrm{de}} N_v \Delta f}\right), \tag{7c}$$

$$\mathcal{S}^u_{\mathrm{do}} = \bigcup_{n_{\mathrm{do}}=0}^{N_{\mathrm{do}}-1} \mathcal{S}^{u,\mathrm{do}}_{n_{\mathrm{do}}},$$
$$\mathcal{S}^{u,\mathrm{do}}_{n_{\mathrm{do}}} = \left[\frac{N_d\left(n_{\mathrm{do}} - \frac{N_{\mathrm{do}}}{2}\right)}{N_{\mathrm{do}} N T_{\mathrm{sym}}} - \nu^{\mathrm{LoS}}_u, \frac{N_d\left(n_{\mathrm{do}} + 1 - \frac{N_{\mathrm{do}}}{2}\right)}{N_{\mathrm{do}} N T_{\mathrm{sym}}} - \nu^{\mathrm{LoS}}_u\right), \tag{7d}$$

where $N_\tau \triangleq \lceil \frac{N_g N_v}{N_c} \rceil$. $N_r = F_r M_r$, $N_t = F_t M_t$, $N_{\mathrm{de}} = F_{\mathrm{de}} N_\tau$ and $N_{\mathrm{do}} = F_{\mathrm{do}} N_d$, where F_r, F_t, F_{de} and F_{do} are the fine factors. Then, (6) can be approximated by in (8), shown at the bottom of the page, where the steering vector $\mathbf{s}\left(\Omega^A_{u,p,q}, \Omega^D_{u,p,q}, \tau_{u,p}, \nu^{\mathrm{cps}}_{u,p}\right)$ for $\Omega^A_{u,p,q} \in \mathcal{S}^r_{n_r}$, $\Omega^D_{u,p,q} \in \mathcal{S}^t_{n_t}$, $\tau_{u,p} \in \mathcal{S}^{\mathrm{de}}_{n_{\mathrm{de}}}$, $\nu^{\mathrm{cps}}_{u,p} \in \mathcal{S}^{u,\mathrm{do}}_{n_{\mathrm{do}}}$ is approximated as the sampled steering vector $\mathbf{s}\left(\Omega^A_{n_r}, \Omega^D_{n_t}, \tau_{n_{\mathrm{de}}}, \nu_{n_{\mathrm{do}}}\right)$ with $\Omega^A_{n_r} = \frac{2n_r - N_r}{N_r}$, $\Omega^D_{n_t} = \frac{2n_t - N_t}{N_t}$, $\tau_{n_{\mathrm{de}}} = \frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}} N_v \Delta f}$ and $\nu_{n_{\mathrm{do}}} = \frac{N_d(n_{\mathrm{do}} - N_{\mathrm{do}}/2)}{N_{\mathrm{do}} N T_{\mathrm{sym}}} - \nu^{\mathrm{LoS}}_u$. As the quantities N_r, N_t, N_{de} and N_{do} of partitioned subsets increase, the accuracy of this approximation improves. We then define a transform matrix $\mathbf{S}_u$ that consists of all sampled steering vectors of UT u with the $(n_{\mathrm{do}} N_{\mathrm{de}} N_t N_r + n_{\mathrm{de}} N_t N_r + n_t N_r + n_r)$-th column being $\mathbf{s}\left(\Omega^A_{n_r}, \Omega^D_{n_t}, \tau_{n_{\mathrm{de}}}, \nu_{n_{\mathrm{do}}}\right)$. $\mathbf{S}_u$ takes the form of

$$\mathbf{S}_u = \mathbf{D}_u \otimes [\mathbf{F} \otimes (\mathbf{V} \otimes \mathbf{U})] \in \mathbb{C}^{NN_vM_tM_r \times N_{\mathrm{do}} N_{\mathrm{de}} N_t N_r}, \tag{9}$$

$$\mathbf{h}^{\mathrm{SFT}}_u = \sum_{n_r=0}^{N_r-1}\sum_{n_t=0}^{N_t-1}\sum_{n_{\mathrm{de}=0}}^{N_{\mathrm{de}-1}}\sum_{n_{\mathrm{do}}=0}^{N_{\mathrm{do}}-1} \sum_{\substack{\Omega^A_{u,p,q} \in \mathcal{S}^r_{n_r}, \Omega^D_{u,p,q} \in \mathcal{S}^t_{n_t}, \\ \tau_{u,p} \in \mathcal{S}^{\mathrm{de}}_{n_{\mathrm{de}}}, \nu^{\mathrm{cps}}_{u,p} \in \mathcal{S}^{u,\mathrm{do}}_{n_{\mathrm{do}}}}} \gamma_{u,p,q} \mathbf{s}\left(\Omega^A_{u,p,q}, \Omega^D_{u,p,q}, \tau_{u,p}, \nu^{\mathrm{cps}}_{u,p}\right), \tag{8}$$

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

where $\mathbf{D}_u \in \mathbb{C}^{N\times N_{\rm do}}$ with $[\mathbf{D}_u]_{ij} = e^{\jmath 2\pi i \nu_j T_{\rm sym}}$, $\mathbf{F} \in \mathbb{C}^{N_v\times N_{\rm de}}$ with $[\mathbf{F}]_{ij} = e^{-\jmath 2\pi (m_0+i)\Delta f \tau_j}$, $\mathbf{V} \in \mathbb{C}^{M_t\times N_t}$ with $[\mathbf{V}]_{ij} = e^{-\jmath 2\pi i \Delta_t \Omega_j^D}$ and $\mathbf{U} \in \mathbb{C}^{M_r\times N_r}$ with $[\mathbf{U}]_{ij} = e^{-\jmath 2\pi i \Delta_r \Omega_j^A}$. In consequence, the SFT domain channel (8) can be expressed by as

$$\mathbf{h}_u^{\rm SFT} = \mathbf{S}_u \mathbf{h}_u^{\rm QB}, \tag{10}$$

where $\mathbf{h}_u^{\rm QB} \in \mathbb{C}^{N_{\rm do}N_{\rm de}N_tN_r\times 1}$ is the QB domain channel and $\mathbf{S}_u$ is the transform matrix from the QB domain channel to the SFT domain channel of UT u. To be specific, $\mathbf{h}_u^{\rm QB} \in \mathbb{C}^{N_{\rm do}N_{\rm de}N_tN_r\times 1}$ is given by in (11), shown at the bottom of the page.

Each sampled steering vector, i.e., each column of $\mathbf{S}_u$, corresponds to a physical quadruple-beam in the SFT domain. Therefore, we refer to (10) as the QB based channel model.

Let $\mathbf{h}^{\rm SFT} = \left(\left(\mathbf{h}_1^{\rm SFT}\right)^H, \left(\mathbf{h}_2^{\rm SFT}\right)^H, \cdots, \left(\mathbf{h}_U^{\rm SFT}\right)^H\right)^H$ denote the stacked channel matrix of all users in the system. $\mathbf{h}^{\rm SFT}$ takes the form of

$$\mathbf{h}^{\rm SFT} = \mathbf{S}\mathbf{h}^{\rm QB} \in \mathbb{C}^{UNN_vM_tM_r\times 1}, \tag{12}$$

where $\mathbf{S} = {\rm blkdiag}\left(\mathbf{S}_1, \cdots, \mathbf{S}_U\right)$ and $\mathbf{h}^{\rm QB} = \left(\left(\mathbf{h}_1^{\rm QB}\right)^H, \left(\mathbf{h}_2^{\rm QB}\right)^H, \cdots, \left(\mathbf{h}_U^{\rm QB}\right)^H\right)^H$. Recall that the viaduct scenario is dominated by the LoS path, indicating that the power of $\mathbf{h}^{\rm QB}$ is dominated by the LoS elements. What's more, the viaduct scenario exhibits limited propagation paths and angle spread due to the sparse scattering environment. All these properties lead to $\mathbf{h}_u^{\rm QB}$ being a sparse vector in massive MIMO HSR communications. Assuming the elements in $\mathbf{h}^{\rm QB}$ follow independent complex Gaussian distributions with zero mean and different variances, the covariance matrix of $\mathbf{h}^{\rm QB}$ is given by

$$\begin{aligned}\mathbf{R}^{\rm QB} &= \mathbb{E}\left\{\mathbf{h}^{\rm QB}\left(\mathbf{h}^{\rm QB}\right)^H\right\} \\ &= {\rm blkdiag}\left(\mathbf{R}_1^{\rm QB}, \cdots, \mathbf{R}_U^{\rm QB}\right),\end{aligned} \tag{13}$$

which is referred to as the statistical CSI in QB domain with $\mathbf{R}_u^{\rm QB}$ being the statistical QB domain CSI of user u. $\mathbf{R}^{\rm QB}$ is a sparse and diagonal matrix and varies relatively slow. We assume that $\mathbf{R}^{\rm QB}$ is known at the BS side in the rest of the paper.

C. Channel Prediction

We consider the pilot-based channel estimation. The sliding window method is widely utilized to integrate the current time slot with the previous time slots into a complete subframe, wherein the pilot segments are assigned for estimation [33], [43]. Therefore, the estimation of the QB domain channel is performed every time a new time slot arrives for a higher accuracy. With a little abuse of definition, the subframe made up of the n_b-th slot and its previous slots is termed as the n_b-th virtual subframe. In this subsection, we focus on the channel prediction during one virtual subframe.

Let

$$\mathbf{S}_{u,\rm p} = \mathbf{D}_{u,\rm p} \otimes \mathbf{F} \otimes \mathbf{V} \otimes \mathbf{U} \tag{14}$$

denote the transform matrix from the QB domain channel to the SFT domain channel at the pilot segments, where $\mathbf{D}_{u,\rm p} = \mathbf{\Theta}_{\rm p}\mathbf{D}_u$ with the n_b-th row of $\mathbf{\Theta} \in \mathbb{C}^{N_b\times N}$ being the $(n_bN_s + n_{\rm p})$-th row of $\mathbf{I}_N$ for $n_b = 1, 2, \cdots, N_b$. The SFT domain channel at pilot segments of the n_b-th virtual subframe can be extracted as

$$\mathbf{h}_{u,\rm p}^{\rm SFT} = \mathbf{S}_{u,\rm p}\mathbf{h}_u^{\rm QB}[n_b] \in \mathbb{C}^{N_bN_vM_tM_r\times 1}. \tag{15}$$

Let $\mathbf{x}_u^{\rm p} \in \mathbb{C}^{N_v\times 1}$ denote the transmitted pilot of UT u at valid subcarriers, $\mathbf{X}_u \triangleq \mathbf{I}_{N_b} \otimes [{\rm diag}\left(\mathbf{x}_u^{\rm p}\right) \otimes \left(\mathbf{I}_{M_t} \otimes \mathbf{I}_{M_r}\right)]$ and $\mathbf{X} \triangleq [\mathbf{X}_0, \mathbf{X}_1, \cdots, \mathbf{X}_{U-1}]$. Then, the received signal $\mathbf{y}^{\rm BS}$ at the BS can be written as

$$\begin{aligned}\mathbf{y}^{\rm BS} &= \sum_{u\in\mathcal{U}} \mathbf{X}_u \mathbf{h}_{u,\rm p}^{\rm SFT} + \mathbf{z}^{\rm BS} \\ &= \mathbf{X}\mathbf{S}^{\rm p}\mathbf{h}^{\rm QB}[n_b] + \mathbf{z}^{\rm BS},\end{aligned} \tag{16}$$

where $\mathbf{S}^{\rm p} = {\rm blkdiag}\left(\mathbf{S}_{1,\rm p}, \cdots, \mathbf{S}_{U,\rm p}\right)$ and $\mathbf{z}^{\rm BS}$ is the Gaussian noise distributed as $\mathcal{CN}\left(0, \sigma_z^2\mathbf{I}_{M_r}\right)$. Note that $\mathbf{h}^{\rm QB}[n_b]$ is sparse and only the non-zero elements need to be estimated in practice. Additionally, numerous statistical inference algorithms are available to estimate $\mathbf{h}^{\rm QB}[n_b]$ effectively [30], [44], [45], [46].

Let $\hat{\mathbf{h}}_u^{\rm QB}[n_b]$ stand for the QB domain channel estimated from the pilot segments in the n_b-th virtual subframe. Then, we use the estimated QB domain channel to predict the channel at the data segment according to (12) and the corresponding estimated SFT domain channel within the n_b-th virtual subframe is given by

$$\hat{\mathbf{h}}_u^{\rm SFT} = \mathbf{S}_u\hat{\mathbf{h}}_u^{\rm QB}[n_b] \in \mathbb{C}^{NN_vM_tM_r\times 1}. \tag{17}$$

With (17), we can obtain the channels of all the OFDM symbols within the n_b-th virtual subframe.

In a quasi-static scenario, it is typical to directly apply the channel estimated from the pilot segments to the rest of symbols for precoding as the channel nearly keeps constant during a time slot. In this case, the estimated SFT domain at the pilot segments can be obtained by setting the selection matrix $\mathbf{S}_u$ in (17) to $\mathbf{S}_{u,\rm p}$ and is given by

$$\hat{\mathbf{h}}_{u,\rm p}^{\rm SFT} = \mathbf{S}_{u,\rm p}\hat{\mathbf{h}}_u^{\rm QB}[n_b] \in \mathbb{C}^{N_bN_vM_tM_r\times 1}, \tag{18}$$

which can be directly employed for precoder design in the quasi-static scenario.

However, the channel changes quickly in HSR communications and will even vary OFDM symbol by OFDM symbol when the moving speed v and center frequency f_c become

$$\left[\mathbf{h}_u^{\rm QB}\right]_{n_{\rm do}N_{\rm de}N_tN_r+n_{\rm de}N_tN_t+n_tN_r+n_r} = \sum_{\substack{\Omega_{u,p,q}^A\in\mathcal{S}_{n_r}^r, \Omega_{u,p,q}^D\in\mathcal{S}_{n_t}^t, \\ \tau_{u,p}\in\mathcal{S}_{n_{\rm de}}^{\rm de}, \nu_{u,p}^{\rm cps}\in\mathcal{S}_{n_{\rm do}}^{u,{\rm do}}}} \gamma_{u,p,q}. \tag{11}$$

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

higher, necessitating the pursuit for a heightened precise CSI at each symbol without increasing the pilot overhead. In this case, we can take advantage of the relationship between $\mathbf{h}_u^{\mathrm{SFT}}$ and $\mathbf{h}_u^{\mathrm{QB}}$ to predict the channel between pilot segments of interest. Let

$$\mathbf{S}_{u,n,m} = \tilde{\mathbf{D}}_{u,n} \otimes \tilde{\mathbf{F}}_m \otimes \mathbf{V} \otimes \mathbf{U} \tag{19}$$

represent the transform matrix from the QB domain channel to the space domain channel at the m-th subcarrier of the n-th OFDM symbol, where $\tilde{\mathbf{D}}_{u,n} = \mathbf{\Theta}_n \mathbf{D}_u \in \mathbb{C}^{1\times N_{\mathrm{do}}}$ with the n-th element of $\mathbf{\Theta}_n \in \mathbb{C}^{1\times N}$ being 1 and the others being 0, and $\tilde{\mathbf{F}}_m = \mathbf{\Phi}_m \mathbf{F} \in \mathbb{C}^{1\times N_{\mathrm{de}}}$ with the m-th element of $\mathbf{\Phi}_m \in \mathbb{C}^{1\times N_v}$ being 1 and the others being 0, respectively. By setting the selection matrix $\mathbf{S}_u$ in (17) to $\mathbf{S}_{u,n,m}$, the predicted channel at the m-th subcarrier of the n-th OFDM symbol can be written as

$$\hat{\mathbf{h}}_{u,m,n}^{\mathrm{S}} = \mathbf{S}_{u,n,m} \hat{\mathbf{h}}_u^{\mathrm{QB}}[n_b] \in \mathbb{C}^{M_t M_r \times 1}. \tag{20}$$

By reshaping $\hat{\mathbf{h}}_{u,m,n}^{\mathrm{S}}$, we can obtain the channel matrix $\hat{\mathbf{H}}_{u,m,n} \in \mathbb{C}^{M_r \times M_t}$ for precoder design at the m-th subcarrier of the n-th OFDM symbol given by

$$\hat{\mathbf{H}}_{u,m,n} = \mathrm{unvec}\left(\mathbf{S}_{u,n,m} \hat{\mathbf{h}}_u^{\mathrm{QB}}[n_b], (M_r, M_t)\right). \tag{21}$$

D. A Posterior QB Based Channel Model

In practice, the accuracy of approximating $\mathbf{h}_{u,m,n}^{\mathrm{S}}$ by $\hat{\mathbf{h}}_{u,m,n}^{\mathrm{S}}$ may suffer from the prediction error caused by the limited resolution ratio in the Doppler domain [30], which is inevitable in the HSR scenario due to the high mobility. Even worse, this inaccuracy could be more severe especially when the symbol n to be predicted is far from the pilot symbol n_{p}. To further improve the performance, we treat the QB domain channels estimated periodically from different virtual subframes as a time series, which facilitates the development of the *a posterior* QB domain channel that better matches the subsequent time slot. Then, the channel between pilot segments can be better predicted.

Recall that the channel parameters are assumed nearly time invariant within a subframe, implying that the QB domain channels estimated from different virtual subframes are distinct and can be treated as a time series anticipated to manifest significant temporal coherence. In addition, as the elements in $\mathbf{h}_u^{\mathrm{QB}}$ are assumed independent of each other, we take the prediction error into consideration and propose *a posterior* QB based channel model to model the temporal coherence of the QB domain channel based on the element-wise Gauss-Markov process [12], [18], [24], [25], [32], given by

$$\mathbf{h}_u^{\mathrm{QB}}[n_b+1] = \hat{\boldsymbol{\beta}}_u \odot \hat{\mathbf{h}}_u^{\mathrm{QB}}[n_b] + \tilde{\boldsymbol{\beta}}_u \odot \left((\mathbf{R}_u^{\mathrm{QB}})^{\frac{1}{2}} \tilde{\mathbf{h}}_u^{\mathrm{QB}}\right), \tag{22}$$

where $\mathbf{h}_u^{\mathrm{QB}}[n_b+1]$ is the *a posterior* QB domain channel during the (n_b+1)-th virtual subframe and $\tilde{\mathbf{h}}_u^{\mathrm{QB}}$ follows the complex Gaussian distribution with i.i.d., zero mean and unit variance entries. $\hat{\boldsymbol{\beta}}_u \in \mathbb{C}^{N_{\mathrm{do}} N_{\mathrm{de}} N_t N_r \times 1}$ reflects the element-wise temporal correlation between the QB domain channels, and can be obtained from the Pearson's linear correlation coefficient. $\tilde{\boldsymbol{\beta}}_u \odot \left((\mathbf{R}_u^{\mathrm{QB}})^{\frac{1}{2}} \tilde{\mathbf{h}}_u^{\mathrm{QB}}\right)$ is the statistical prediction error with

$$|(\tilde{\boldsymbol{\beta}}_u)_i|^2 = 1 - |(\hat{\boldsymbol{\beta}}_u)_i|^2. \tag{23}$$

Then, (22) can be used to predict the *a posterior* space domain channel for precoding. Let

$$\mathbf{S}_{u,n,m}^{\mathrm{A}} = \tilde{\mathbf{D}}_{u,n} \otimes \tilde{\mathbf{F}}_m \otimes \mathbf{I}_{N_t} \otimes \mathbf{I}_{N_r} \tag{24}$$

denote the transform matrix from the QB domain channel to the angle domain channel at the m-th subcarrier of the n-th OFDM symbol. For the OFDM symbol n that lies between the pilot segments of the n_b-th and (n_b+1)-th slots, the corresponding *a posterior* angle domain channel vector of user u at the m-th subcarrier of the n-th OFDM symbol can be expressed as

$$\begin{aligned}\mathbf{h}_{u,m,n}^{\mathrm{A}} &= \mathbf{S}_{u,n,m}^{\mathrm{A}} \mathbf{h}_u^{\mathrm{QB}}[n_b+1] \\ &= \hat{\mathbf{h}}_{u,n,m}^{\mathrm{A}} + \tilde{\mathbf{h}}_{u,m,n}^{\mathrm{A}},\end{aligned} \tag{25}$$

where $\hat{\mathbf{h}}_{u,n,m}^{\mathrm{A}} = \mathbf{S}_{u,n,m}^{\mathrm{A}}(\hat{\boldsymbol{\beta}}_u \odot \hat{\mathbf{h}}_u^{\mathrm{QB}}[n_b])$, $\tilde{\mathbf{h}}_{u,m,n}^{\mathrm{A}} = \mathbf{S}_{u,n,m}^{\mathrm{A}}\left(\tilde{\boldsymbol{\beta}}_u \odot ((\mathbf{R}_u^{\mathrm{QB}})^{\frac{1}{2}} \tilde{\mathbf{h}}_u^{\mathrm{QB}})\right)$ are the deterministic and statistical components, respectively. $\tilde{\mathbf{h}}_{u,m,n}^{\mathrm{A}}$ is a complex Gaussian random vector with zero mean and covariance matrix

$$\tilde{\mathbf{R}}_{u,n,m}^{\mathrm{A}} = \mathbf{S}_{u,n,m}^{\mathrm{A}} \tilde{\mathbf{R}}_u^{\mathrm{QB}} \left(\mathbf{S}_{u,n,m}^{\mathrm{A}}\right)^H, \tag{26}$$

where

$$\begin{aligned}\tilde{\mathbf{R}}_u^{\mathrm{QB}} &= \mathrm{diag}\left(\tilde{\boldsymbol{\beta}}_u \odot \tilde{\boldsymbol{\beta}}_u^*\right) \mathbf{R}_u^{\mathrm{QB}} \\ &= \left(\mathbf{I} - \mathrm{diag}\left(\hat{\boldsymbol{\beta}}_u \odot \hat{\boldsymbol{\beta}}_u^*\right)\right) \mathbf{R}_u^{\mathrm{QB}}.\end{aligned} \tag{27}$$

It is easy to verify that $\tilde{\mathbf{R}}_{u,n,m}^{\mathrm{A}}$ is still a diagonal matrix. Correspondingly, $\mathbf{h}_{u,m,n}^{\mathrm{A}}$ is a complex Gaussian random vector characterized by its mean $\hat{\mathbf{h}}_{u,n,m}^{\mathrm{A}}$, covariance matrix $\tilde{\mathbf{R}}_{u,n,m}^{\mathrm{A}}$ and autocorrelation matrix

$$\mathbf{R}_{u,n,m}^{\mathrm{A}} = \hat{\mathbf{h}}_{u,n,m}^{\mathrm{A}} \left(\hat{\mathbf{h}}_{u,n,m}^{\mathrm{A}}\right)^H + \tilde{\mathbf{R}}_{u,n,m}^{\mathrm{A}}. \tag{28}$$

With the *a posterior* angle domain channel, the *a posterior* space domain channel vector takes the form of

$$\mathbf{h}_{u,n,m}^{\mathrm{S}} = (\mathbf{V} \otimes \mathbf{U})\, \mathbf{h}_{u,n,m}^{\mathrm{A}} \tag{29}$$

with the autocorrelation matrix given by

$$\mathbf{R}_{u,n,m}^{\mathrm{S}} = \hat{\mathbf{h}}_{u,n,m}^{\mathrm{A}} \left(\hat{\mathbf{h}}_{u,n,m}^{\mathrm{A}}\right)^H + \tilde{\mathbf{R}}_{u,n,m}^{\mathrm{A}}. \tag{30}$$

Further, the *a posterior* space domain channel matrix at the m-th subcarrier of the n-th OFDM symbol is given by

$$\begin{aligned}\mathbf{H}_{u,m,n} &= \mathrm{unvec}\left(\mathbf{h}_{u,n,m}^{\mathrm{S}}, (M_r, M_t)\right) \\ &\overset{(a)}{=} \mathbf{U}\left(\mathrm{unvec}\left(\hat{\mathbf{h}}_{u,n,m}^{\mathrm{A}} + \tilde{\mathbf{h}}_{u,m,n}^{\mathrm{A}}, (N_r, N_t)\right)\right)\mathbf{V}^T \\ &= \hat{\mathbf{H}}_{u,m,n} + \mathbf{U}\tilde{\mathbf{G}}_{u,n,m}\mathbf{V}^T,\end{aligned} \tag{31}$$

where (a) follows from $\left(\mathbf{B}^T \otimes \mathbf{A}\right)\mathrm{vec}(\mathbf{X}) = \mathrm{vec}(\mathbf{AXB})$. $\tilde{\mathbf{G}}_{u,n,m} = \mathrm{unvec}\left(\tilde{\mathbf{h}}_{u,m,n}^{\mathrm{A}}, (N_r, N_t)\right)$ is the matrix form

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

of $\tilde{\mathbf{h}}^{\mathrm{A}}_{u,m,n}$ with

$$\tilde{\mathbf{\Omega}}_{u,n,m} \triangleq \mathbb{E}\left\{\tilde{\mathbf{G}}_{u,n,m} \odot \tilde{\mathbf{G}}^{*}_{u,n,m}\right\} = \mathrm{unvec}\left(\mathrm{diag}\left(\tilde{\mathbf{R}}^{\mathrm{A}}_{u,n,m}\right), (N_r, N_t)\right). \quad (32)$$

$\tilde{\mathbf{\Omega}}_{u,n,m}$ is referred to as the angle domain power matrix.

By setting $\hat{\boldsymbol{\beta}}_u$ according the time interval between the current pilot segment and the forthcoming virtual subframe to be predicted, we are able to describe the channel uncertainties in various channel conditions. If $\hat{\boldsymbol{\beta}}_u = \mathbf{1}$, the *a posterior* space domain channel (29) is reduced to the predicted channel (20). If $\hat{\boldsymbol{\beta}}_u = \mathbf{0}$, the precoders are designed only with the statistical CSI. In fact, the *a posterior* channel model enables predictions over longer time duration by adjusting $\hat{\boldsymbol{\beta}}_u$, which can efficiently conserve the pilot resources. Next we will investigate the manifold optimization-based robust precoder design along with the proposed *a posterior* space domain channel (31).

III. Problem Formulation and Riemannian Ingredients in Matrix Manifold Optimization

A. Problem Formulation on the Riemannian Submanifold

We investigate the robust precoder design against channel aging and prediction error in HSR scenario at a given sub-carrier, whose subscript is thereby omitted for brevity. Let $\mathbf{H}_{kn}$ denote the channel of UT k at a given sub-carrier of the n-th OFDM symbol. Let $\mathbf{x}_{kn} \in \mathbb{C}^{d_k}$ denote the signal transmitted to the k-th UT at the given subcarrier of the n-th symbol satisfying $\mathbb{E}\left\{\mathbf{x}_{kn}\mathbf{x}^H_{kn}\right\} = \mathbf{I}_{d_k}$. The received signal of the k-th UT at the n-th OFDM symbol is given by

$$\mathbf{y}_{kn} = \overbrace{\mathbf{H}_{kn}\mathbf{P}_{kn}\mathbf{x}_{kn}}^{\text{desired signal}} + \overbrace{\mathbf{H}_{kn}\sum_{\ell\neq i,\ell=1}^{U}\mathbf{P}_{\ell n}\mathbf{x}_{\ell n} + \mathbf{z}_{kn}}^{\text{interference plus noise}}, \quad (33)$$

where $\mathbf{P}_{kn} \in \mathbb{C}^{M_t \times d_k}$ is the precoding matrix of user k at the n-th OFDM symbol and $\mathbf{z}_{kn}$ is the Gaussian noise distributed as $\mathcal{CN}\left(0, \sigma_z^2\mathbf{I}_{M_r}\right)$. The interference plus noise is treated as the Gaussian noise for the worst case design with the covariance matrix given by

$$\mathbf{R}_{kn} = \mathbb{E}_{\mathbf{H}_{kn}}\left\{\mathbf{H}_{kn}\sum_{\ell\neq k}\mathbf{P}_{\ell n}\mathbf{P}^H_{\ell n}\mathbf{H}^H_{kn}\right\} + \sigma_z^2\mathbf{I}_{M_r}. \quad (34)$$

Assuming that $\mathbf{R}_{kn}$ is known at the user side, the ergodic rate of user k can be written as

$$\mathcal{R}_{kn} = \mathbb{E}_{\mathbf{H}_{kn}}\left\{\log\det\left(\mathbf{H}_{kn}\mathbf{P}_{kn}\mathbf{P}^H_{kn}\mathbf{H}^H_{kn} + \mathbf{R}_{kn}\right)\right\} - \log\det\left(\mathbf{R}_{kn}\right), \quad (35)$$

and the ergodic weighted sum-rate maximization problem can be formulated as

$$\mathcal{P}_1: \arg\max_{\mathbf{P}_n}\sum_{k\in\mathcal{U}} w_k\mathcal{R}_{kn} \quad \text{s.t.} \sum_{k\in\mathcal{U}}\mathrm{tr}\left(\mathbf{P}^H_{kn}\mathbf{P}_{kn}\right) = P_t, \quad (36)$$

where P_t is the total power constraint. However, $\mathcal{R}_{kn}$ has no closed-form due to the expectation operator, making it difficult to directly tackle with (36) [47]. Note that $\mathrm{logdet}\left(\mathbf{A}\right)$ is concave when $\mathbf{A}$ is a positive-definite matrix. So, the Jensen's inequality can be applied to (35), leading to an upper bound given by

$$\mathcal{R}_{kn} \leq \mathcal{R}^{\mathrm{ub}}_{kn} = \mathrm{logdet}\{\mathbf{P}^H_{kn}\mathbb{E}\left\{\mathbf{H}^H_{kn}\mathbf{R}^{-1}_{kn}\mathbf{H}_{kn}\right\}\mathbf{P}_{kn} + \mathbf{I}_{d_i}\}. \quad (37)$$

With (37), the optimization problem can be reformulated as

$$\mathcal{P}_2: \arg\max_{\mathbf{P}_n}\sum_{k\in\mathcal{U}} w_k\mathcal{R}^{\mathrm{ub}}_{kn} \quad \text{s.t.} \sum_{k\in\mathcal{U}}\mathrm{tr}\left(\mathbf{P}^H_{kn}\mathbf{P}_{kn}\right) = P_t. \quad (38)$$

Conventionally, (38) can be directly solved by virtue of the Lagrange multiplier method [25], which can be reformulated as an unconstrained problem by introducing the Lagrangian multiplier. The bisection method is commonly utilized to search for the Lagrangian multiplier, which involves the inverses of matrices of transmit antenna dimension during the iteration and has a high computational complexity [18]. To avoid the inverses of large dimensional matrices and mitigate the computational complexities, it is proposed to solve the problem on the manifold [38]. Let $\mathbf{P}_n = [\mathbf{P}_{1n}, \cdots, \mathbf{P}_{Un}] \in \mathbb{C}^{M_t \times \sum_{k\in\mathcal{U}} d_k}$ denote the stacked precoding matrix. It has been proved in [38, Theorem 1] that the set

$$\mathcal{M} = \left\{\mathbf{P}_n \mid F\left(\mathbf{P}_n\right) = \mathrm{tr}\left(\mathbf{P}^H_n\mathbf{P}_n\right) - P_t = 0\right\} \quad (39)$$

forms a Riemannian submanifold of the linear manifold $\mathcal{N} := \mathbb{C}^{M_t \times \sum_{k\in\mathcal{U}} d_k}$ with the Riemannian metric

$$g\left(\boldsymbol{\xi}_{\mathbf{P}_n}, \boldsymbol{\zeta}_{\mathbf{P}_n}\right) = \Re\left\{\mathrm{tr}\left(\boldsymbol{\xi}^H_{\mathbf{P}_n}\boldsymbol{\zeta}_{\mathbf{P}_n}\right)\right\}, \quad (40)$$

where $\boldsymbol{\xi}_{\mathbf{P}_n}$ and $\boldsymbol{\zeta}_{\mathbf{P}_n}$ are two tangent vectors to $\mathcal{M}$ at $\mathbf{P}_n$. Therefore, the constrained problem (38) can also be reformulated as an unconstrained one on the Riemannian submanifold $\mathcal{M}$ as

$$\mathcal{P}_3: \arg\min_{\mathbf{P}_n\in\mathcal{M}} f\left(\mathbf{P}_n\right), \quad (41)$$

where $f\left(\mathbf{P}_n\right) = -\sum_{k\in\mathcal{U}} w_k\mathcal{R}^{\mathrm{ub}}_{kn}$ is the objective function.

B. Riemannian Ingredients for Robust Precoder Design

Every point $\mathbf{P}_n$ on the nonlinear $\mathcal{M}$ is attached to a linear tangent space denoted by $T_{\mathbf{P}_n}\mathcal{M}$, which is a set formed by all the tangent vectors to $\mathcal{M}$ at $\mathbf{P}_n$. By introducing the retraction, which is a mapping from the tangent space to the manifold, we can generalize the notion of moving along the tangent vector while remaining on the nonlinear manifold $\mathcal{M}$. The Riemannian submanifold $\mathcal{M}$ is a sphere, whose retraction is given by

$$R^{\mathcal{M}}_{\mathbf{P}_n}\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right) = \frac{\sqrt{P_t}\left(\mathbf{P}_n + \boldsymbol{\xi}_{\mathbf{P}_n}\right)}{\sqrt{\mathrm{tr}\left(\left(\mathbf{P}_n + \boldsymbol{\xi}_{\mathbf{P}_n}\right)^H\left(\mathbf{P}_n + \boldsymbol{\xi}_{\mathbf{P}_n}\right)\right)}} = \rho\left(\mathbf{P}_n + \boldsymbol{\xi}_{\mathbf{P}_n}\right). \quad (42)$$

With the retraction, the objective function f defined in a neighborhood of $\mathbf{P}_n \in \mathcal{M}$ can be transformed into a function defined on the linear tangent space $T_{\mathbf{P}_n}\mathcal{M}$ and can be expressed as [48]

$$h\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right) = f\left(R^{\mathcal{M}}_{\mathbf{P}_n}\left(\boldsymbol{\xi}_{\mathbf{P}_n}\right)\right) \quad (43)$$

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

with $h(\mathbf{0}) = f(\mathbf{P}_n)$. Instead of directly optimizing $\mathbf{P}_n$ on the nonlinear manifold $\mathcal{M}$, (43) paves the way for searching for a good point on the linear tangent space, which facilitates the optimization on the nonlinear Riemannian manifold $\mathcal{M}$. The direction of an efficient $\boldsymbol{\xi}_{\mathbf{P}_n}$ is usually related to the negative Riemannian gradient. Let

$$\check{\mathbf{R}}_{kn} = \mathbb{E}_{\mathbf{H}_{kn}}\left\{\mathbf{H}_{kn}\sum_{\ell\in\mathcal{U}}\mathbf{P}_{\ell n}\mathbf{P}_{\ell n}^H\mathbf{H}_{kn}^H\right\} + \sigma_z^2\mathbf{I}_{M_r}, \tag{44}$$

we have the following theorem.

Theorem 1: The Euclidean gradient of the $f(\mathbf{P}_n)$ is given by

$$\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_n) = \left(\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_{1n}), \cdots, \mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_{Un})\right), \tag{45}$$

where

$$\begin{aligned}\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_{kn}) = &-2\sum_{\ell\in\mathcal{U}}\mathbb{E}_{\mathbf{H}_{\ell n}}\left\{\mathbf{H}_{\ell n}^H\check{\mathbf{R}}_{\ell n}^{-1}\mathbf{H}_{\ell n}\right\}\mathbf{P}_{kn}\\ &+2\sum_{\ell\neq k}\mathbb{E}_{\mathbf{H}_{\ell n}}\left\{\mathbf{H}_{\ell n}^H\mathbf{R}_{\ell n}^{-1}\mathbf{H}_{\ell n}\right\}\mathbf{P}_{kn}.\end{aligned} \tag{46}$$

The Riemannian gradient of $f(\mathbf{P}_n)$ on $T_{\mathbf{P}_n}\mathcal{M}$ is given by

$$\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n) = \mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_n) - \lambda\mathbf{P}_n, \tag{47}$$

where

$$\lambda = \frac{1}{P}\Re\left\{\mathrm{tr}\left(\mathbf{P}_n^H\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_n)\right)\right\}. \tag{48}$$

Proof: The proof is given in Appendix A. □

To accelerate the convergence speed, it is suggested that the previous update direction should be taken into consideration [48]. Let $\boldsymbol{\xi}_{\mathbf{P}_n} \in T_{\mathbf{P}_n}\mathcal{M}$ and $\boldsymbol{\zeta}_{R_{\mathbf{P}_n}^{\mathcal{M}}(\boldsymbol{\eta}_{\mathbf{P}_n})} \in T_{R_{\mathbf{P}_n}^{\mathcal{M}}(\boldsymbol{\eta}_{\mathbf{P}_n})}\mathcal{M}$ denote two tangent vectors in different tangent spaces. It is illegal to directly sum $\boldsymbol{\xi}_{\mathbf{P}_n}$ and $\boldsymbol{\zeta}_{\mathbf{P}_n}$ in different tangent spaces due to the nonlinearity of $\mathcal{M}$. The vector transport denoted by $\mathcal{T}_{\boldsymbol{\eta}_{\mathbf{P}_n}}^{\mathcal{M}}(\boldsymbol{\xi}_{\mathbf{P}_n})$ can transport the tangent vector $\boldsymbol{\xi}_{\mathbf{P}_n}$ from $T_{\mathbf{P}_n}\mathcal{M}$ to another tangent space $T_{R_{\mathbf{P}_n}^{\mathcal{M}}(\boldsymbol{\eta}_{\mathbf{P}_n})}\mathcal{M}$, facilitating the operations between the tangent vectors in different tangent spaces. For the Riemannian submanifold $\mathcal{M}$, $\mathcal{T}_{\boldsymbol{\eta}_{\mathbf{P}_n}}^{\mathcal{M}}(\boldsymbol{\xi}_{\mathbf{P}_n})$ can be achieved by orthogonally projecting $\boldsymbol{\xi}_{\mathbf{P}_n}$ onto $T_{R_{\mathbf{P}_n}^{\mathcal{M}}(\boldsymbol{\eta}_{\mathbf{P}_n})}\mathcal{M}$. Let $\mathbf{P}_n^{\mathrm{new}} = R_{\mathbf{P}_n}^{\mathcal{M}}(\boldsymbol{\eta}_{\mathbf{P}_n})$, the vector transport takes the form of

$$\begin{aligned}\mathcal{T}_{\boldsymbol{\eta}_{\mathbf{P}_n}}^{\mathcal{M}}(\boldsymbol{\xi}_{\mathbf{P}_n}) &= \boldsymbol{\Pi}_{T_{\mathbf{P}_n^{\mathrm{new}}}\mathcal{M}}^{T_{\mathbf{P}_n}\mathcal{M}}(\boldsymbol{\xi}_{\mathbf{P}_n})\\ &= \boldsymbol{\xi}_{\mathbf{P}_n} - \varphi\mathbf{P}_n^{\mathrm{new}},\end{aligned} \tag{49}$$

where

$$\varphi = \frac{1}{P}\Re\left\{\mathrm{tr}\left((\mathbf{P}_n^{\mathrm{new}})^H\boldsymbol{\xi}_{\mathbf{P}_n}\right)\right\}, \tag{50}$$

and $\boldsymbol{\Pi}_{T_{\mathbf{P}}\mathcal{M}}^{T_{\mathbf{P}}\mathcal{N}}(\boldsymbol{\xi}_{\mathbf{P}})$ denotes the orthogonal projection of the tangent vector $\boldsymbol{\xi}_{\mathbf{P}}$ from $T_{\mathbf{P}}\mathcal{N}$ to $T_{\mathbf{P}}\mathcal{M}$.

IV. Riemannian Conjugate Gradient Method for Robust Precoder Design

A. Riemannian Conjugate Gradient Method

In this subsection, we propose to solve (41) with the RCG method.

For the optimization on the nonlinear $\mathcal{M}$, the retraction (42) is utilized to transform the function f defined on $\mathcal{M}$ into the function $\mathrm{h}(\boldsymbol{\xi}_{\mathbf{P}_n})$ defined on the linear tangent space, where $\boldsymbol{\xi}_{\mathbf{P}_n} = \alpha\boldsymbol{\eta}_n$ with $\boldsymbol{\eta}_n$ and α being the search direction and step length, respectively. For notational clarity, any notation with the superscript p denotes the notation in the p-th outer iteration, and any notation with the superscript pair (p,q) represents the notation in the q-th iteration of searching for the step length during the p-th outer iteration. With the search direction $\boldsymbol{\eta}_n^p$, the step length α^p needs to be searched along the search direction for an effective point, which can be effectively achieved via the backtracking method [48]. The update formula when searching for the step length on $\mathcal{M}$ is given by

$$\mathbf{P}_n^{p,q} = \rho^{p,q}\left(\mathbf{P}_n^p + \alpha^{p,q}\boldsymbol{\eta}_n^p\right), \tag{51}$$

where $\alpha^{p,q}$ is the trial step length in the (p,q)-th iteration. (51) is repeated until a satisfactory $\alpha^p = \alpha^{p,q}$ is obtained with $\mathbf{P}_n^{p+1} = \mathbf{P}_n^{p,q+1}$ being the input of the next outer iteration. During this process, the objective function needs to be compared repeatedly until a sufficient decrease is obtained. The objective function in the (p,q)-th iteration can be rewritten as

$$f(\mathbf{P}_n^{p,q}) = -\sum_{k\in\mathcal{U}} w_k\mathcal{R}_{kn}^{p,q}, \tag{52}$$

where

$$\mathcal{R}_{kn}^{p,q} = \mathrm{logdet}\left(\check{\mathbf{R}}_{kn}^{p,q}\right) - \mathrm{logdet}\left(\mathbf{R}_{kn}^{p,q}\right). \tag{53}$$

For a fast convergence speed, the search direction $\boldsymbol{\eta}_n^p$ is chosen to be the Riemannian conjugate gradient and can be expressed as

$$\boldsymbol{\eta}_n^p = -\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p) + \beta^{p-1}\mathcal{T}_{\alpha^{p-1}\boldsymbol{\eta}_n^{p-1}}\left(\boldsymbol{\eta}_n^{p-1}\right). \tag{54}$$

In (54), $\beta^p \in \mathbb{R}$ is the RCG update parameter with several alternatives that yield different nonlinear RCG methods. β^p is chosen as the modified Polak and Ribière parameter (PRP) to avoid jamming [49] and is given by

$$\beta^p = \max\left(0, \min\left(\beta_{\mathrm{PRP}}^p, \beta_{\mathrm{FR}}^p\right)\right), \tag{55}$$

where

$$\beta_{\mathrm{FR}}^p = \frac{g_{\mathbf{P}_n^p}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p), \mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p)\right)}{g_{\mathbf{P}_n^{p-1}}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f\left(\mathbf{P}_n^{p-1}\right), \mathrm{grad}_{\mathcal{M}} f\left(\mathbf{P}_n^{p-1}\right)\right)}. \tag{56}$$

Let $\boldsymbol{\nu}_n^p = \mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p) - \mathcal{T}_{\alpha^{p-1}\boldsymbol{\eta}_n^{p-1}}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f\left(\mathbf{P}^{p-1}\right)\right)$, β_{PRP}^p is given by

$$\beta_{\mathrm{PRP}}^p = \frac{g_{\mathbf{P}^p}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f(\mathbf{P}_n^p), \boldsymbol{\nu}_n^p\right)}{g_{\mathbf{P}_n^{p-1}}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}} f\left(\mathbf{P}_n^{p-1}\right), \mathrm{grad}_{\mathcal{M}} f\left(\mathbf{P}_n^{p-1}\right)\right)}. \tag{57}$$

B. Efficient Algorithm Implementation

With the search direction (54), the problem (41) can be directly solved by the conventional RCG method [50]. Nevertheless, the objective function (52) and the search direction (54) both call for the expectation operator, which could introduce much computational complexity if the Monte-Carlo method is utilized [24]. To be specific, it is that $\mathbf{R}_{kn}^{p,q}$, $\check{\mathbf{R}}_{kn}^{p,q}$ and Euclidean gradient $\mathrm{grad}_{\mathcal{N}} f(\mathbf{P}_n^p)$ need repeatedly calls for the expectation operators. By utilizing the previously derived

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

predicted covariance matrix $\tilde{\mathbf{R}}_{kn}^{\mathrm{A}}$ (26) and the angle domain power matrix $\tilde{\mathbf{\Omega}}_{kn}$ in (32), the Monte-Carlo method can be skillfully avoided to reduce the computational complexity.

With (31), we have

$$\begin{aligned}\mathbb{E}_{\mathbf{H}_{kn}}\left\{\mathbf{H}_{kn}^H\mathbf{A}\mathbf{H}_{kn}\right\}&=\hat{\mathbf{H}}_{kn}^H\mathbf{A}\hat{\mathbf{H}}_{kn}+\mathbb{E}_{\tilde{\mathbf{H}}_{kn}}\left\{\tilde{\mathbf{H}}_{kn}^H\mathbf{A}\tilde{\mathbf{H}}_{kn}\right\}\\&=\hat{\mathbf{H}}_{kn}^H\mathbf{A}\hat{\mathbf{H}}_{kn}+\mathbf{V}^*\mathrm{diag}\left(\tilde{\mathbf{\Omega}}_{kn}^T\mathrm{diag}\left(\mathbf{U}^H\mathbf{A}\mathbf{U}\right)\right)\mathbf{V}^T\\&=\hat{\mathbf{H}}_{kn}^H\mathbf{A}\hat{\mathbf{H}}_{kn}+\mathbf{\Psi}_{kn}\left(\mathbf{A}\right)\end{aligned}\tag{58}$$

and

$$\begin{aligned}\mathbb{E}_{\mathbf{H}_{kn}}\left\{\mathbf{H}_{kn}\mathbf{B}\mathbf{H}_{kn}^H\right\}&=\hat{\mathbf{H}}_{kn}\mathbf{B}\hat{\mathbf{H}}_{kn}^H+\mathbb{E}_{\tilde{\mathbf{H}}_{kn}}\left\{\tilde{\mathbf{H}}_{kn}\mathbf{B}\tilde{\mathbf{H}}_{kn}^H\right\}\\&=\hat{\mathbf{H}}_{kn}\mathbf{B}\hat{\mathbf{H}}_{kn}^H+\mathbf{U}\mathrm{diag}\left(\tilde{\mathbf{\Omega}}_{kn}\mathrm{diag}\left(\mathbf{V}^T\mathbf{B}\mathbf{V}^*\right)\right)\mathbf{U}^H\\&=\hat{\mathbf{H}}_{kn}\mathbf{B}\hat{\mathbf{H}}_{kn}^H+\mathbf{\Phi}_{kn}\left(\mathbf{B}\right),\end{aligned}\tag{59}$$

where $\mathbf{\Psi}_{kn}\left(\mathbf{A}\right)\triangleq\mathbf{V}^*\mathrm{diag}\left(\tilde{\mathbf{\Omega}}_{kn}^T\mathrm{diag}\left(\mathbf{U}^H\mathbf{A}\mathbf{U}\right)\right)\mathbf{V}^T$ and $\mathbf{\Phi}_{kn}\left(\mathbf{B}\right)\triangleq\mathbf{U}\mathrm{diag}\left(\tilde{\mathbf{\Omega}}_{kn}\mathrm{diag}\left(\mathbf{V}^T\mathbf{B}\mathbf{V}^*\right)\right)\mathbf{U}^H$.

Note that $\mathbf{U}$ and $\mathbf{V}$ are partial DFT matrices, indicating that (58) and (59) can be effectively computed. With (58) and (59), the expectation operators in $\mathbf{R}_{kn}^{p,q}$, $\check{\mathbf{R}}_{kn}^{p,q}$ and $\mathrm{grad}_{\mathcal{N}}f\left(\mathbf{P}_n^p\right)$ can be efficiently obtained without relying on the Monte-Carlo method.

However, $\mathbf{R}_{kn}^{p,q}$ and $\check{\mathbf{R}}_{kn}^{p,q}$ still need to be calculated repeatedly during the inner iteration for comparing the value of the objective function (52), implying that the computational complexity is dependent on the number of inner iterations, which is still unacceptable in practice. For clarity, we summarize the detailed derivations in Appendix B, which shows that the computational complexity can be significantly decreased and is irrelevant to the number of inner iterations by precomputing some elements.

The RCG method for robust precoder design is provided in Algorithm 1, where r and c are typically chosen as 0.5 and 10^{-4}, respectively. As the search direction is determined first in the RCG method, and then a limited number of trail steps are generated in order to find a point with a sufficient decrease of the cost function along the search direction. Therefore, the proposed method is monotonic [38]. If the precoder is designed only with the predicted channel (21) without considering the prediction error, the input $\tilde{\mathbf{\Omega}}_{kn}$ can be directly set to $\mathbf{0}$ and the RCG method is reduced to the RCG method proposed in [38].

C. Computational Complexity

The computational complexities of the proposed RCG method are linear with the number of outer iterations, and nearly independent of the number of inner iterations. Let N_{out}, $N_S=\sum_{k\in\mathcal{U}}d_k$, and $N_R=UN_r$ denote the total numbers of the outer iterations, data streams and user antennas, respectively. Then, we focus on the complexity analysis of each iteration.

For ease of analysis, we first provide the computational complexities of some Riemannian ingredients and operators, which will be repeatedly invoked during the iteration. The Riemannian metric, orthogonal projection, retraction and vector transport are computationally efficient and share the same

Algorithm 1 RCG Method for Robust Precoder Design

Require: Riemannian submanifold $\mathcal{M}$; Riemannian metric $g_{\mathbf{P}}^{\mathcal{M}}\left(\cdot\right)$; Real-valued function f; Retraction $R_{\mathbf{P}}^{\mathcal{M}}\left(\cdot\right)$; Vector transport $\mathcal{T}_{\boldsymbol{\eta}_n}^{\mathcal{M}}\left(\cdot\right)$; initial step length $\alpha^0>0$; $r\in(0,1)$; $c\in(0,1)$

Input: $\hat{\mathbf{H}}_{kn}$, $\tilde{\mathbf{\Omega}}_{kn},\forall k\in\mathcal{U}$; initial point $\mathbf{P}^0$;

repeat
  Get $\hat{\mathbf{E}}_{k\ell}^p=\hat{\mathbf{H}}_{kn}\mathbf{P}_{\ell n}^p$ for $k,\ell\in\mathcal{U}$.
  Compute Euclidean gradient $\mathrm{grad}_{\mathcal{N}}f(\mathbf{P}_{kn}^p)$ with (74).
  Get Riemannian gradient $\mathrm{grad}_{\mathcal{M}}f(\mathbf{P}_{kn}^p)$ with (47).
  Update the search direction $\boldsymbol{\eta}_n^p$ with (54) and compute $\bar{\mathbf{E}}_{k\ell}^p=\hat{\mathbf{H}}_{kn}\boldsymbol{\eta}_{\ell n}^p,k,\ell\in\mathcal{U}$.
  Compute $\hat{\mathbf{F}}_{k\ell}^p$, $\bar{\mathbf{F}}_{k\ell}^p$ and $\tilde{\mathbf{F}}_{k\ell}^p$.
  while $f\left(\mathbf{P}_n^{p,q-1}\right)-f\left(\mathbf{P}_n^p\right)\geq c\times g_{\mathbf{P}^n}^{\mathcal{M}}\left(\mathrm{grad}_{\mathcal{M}}f\left(\mathbf{P}_n^p\right),\alpha^{n,m-1}\boldsymbol{\eta}_n^p\right)$ **do**
    Set $\alpha^{p,q}\leftarrow r\alpha^{p,q-1}$ with $\alpha^{p,0}=\alpha^0$.
    Get $\mathbf{P}_n^{p,q+1}$ with (51) and $\hat{\mathbf{E}}_{k\ell}^{p,q},i,j\in\mathcal{S}_U$ with (71).
    Get $f\left(\mathbf{P}_n^{p,q}\right)$ with (52), (72) and (73), $q\leftarrow q+1$.
  end while
  Set $\mathbf{P}_n^{p+1}\leftarrow\mathbf{P}_n^{p,q}$, $f\left(\mathbf{P}_n^{p+1}\right)\leftarrow f\left(\mathbf{P}_n^{p,q}\right)$, $\hat{\mathbf{E}}_{k\ell}^{p+1}\leftarrow\hat{\mathbf{E}}_{k\ell}^{p,q}$, $\hat{\mathbf{F}}_{k\ell}^{p+1}\leftarrow\mathbf{\Phi}_{kn}\left(\mathbf{P}_{\ell n}^{p,q}\left(\left(\mathbf{P}_{\ell n}^{p,q}\right)^H\right)\right)$, $\mathbf{R}_{kn}^{p+1}\leftarrow\mathbf{R}_{kn}^{p,q}$ and $\check{\mathbf{R}}_{kn}^{p+1}\leftarrow\check{\mathbf{R}}_{kn}^{p,q},\forall k,\ell\in\mathcal{U}$, $p\leftarrow p+1$.
until convergence

TABLE I
THE COMPUTATIONAL COMPLEXITIES OF THE INGREDIENTS USED IN ALGORITHM 1

Ingredients	Computational complexity
$g_{\mathbf{P}}^{\mathcal{M}}\left(\boldsymbol{\xi}_{\mathbf{P}},\boldsymbol{\zeta}_{\mathbf{P}}\right)$	M_tN_S
$\mathbf{\Pi}_{T_{\mathbf{P}}\mathcal{M}}^{T_{\mathbf{P}}\mathcal{N}}\left(\boldsymbol{\xi}_{\mathbf{P}}\right)$	M_tN_S
$R_{\mathbf{P}}^{\mathcal{M}}\left(\boldsymbol{\xi}_{\mathbf{P}}\right)$	M_tN_S
$\mathcal{T}_{\boldsymbol{\eta}_{\mathbf{P}}}^{\mathcal{M}}\left(\boldsymbol{\xi}_{\mathbf{P}}\right)$	M_tN_S
$\bar{\mathbf{E}}_{k\ell},\forall k,\ell\in\mathcal{U}$	$M_tN_SN_R$
$\bar{\mathbf{F}}_{k\ell},\tilde{\mathbf{F}}_{k\ell},k,\ell\in\mathcal{U}$	$O(2M_t\log(N_t)N_S+N_t^2)$
$\mathrm{grad}_{\mathcal{N}}f\left(\mathbf{P}\right)$	$O(M_tN_SN_R+M_t^2\log(N_t)+M_t^2N_S)$

computational complexity of M_tN_s. $\mathbf{\Phi}_{kn}\left(\cdot\right)$ and $\mathbf{\Psi}_{kn}\left(\cdot\right)$ can be realized via FFT operations. To be specific, $\mathbf{\Phi}_{kn}\left(\cdot\right)$ is invoked once per outer iteration for all users by $\bar{\mathbf{F}}_{k\ell}^p$ and $\tilde{\mathbf{F}}_{k\ell}^p$ defined in (70). $\mathbf{\Psi}_{kn}\left(\cdot\right)$ is invoked once per outer iteration for all users by $\mathrm{grad}_{\mathcal{N}}f\left(\mathbf{P}_{kn}^{p+1}\right)$ defined in (74). With $\mathbf{\Phi}_{kn}\left(\cdot\right)$ and $\mathbf{\Psi}_{kn}\left(\cdot\right)$, we can further compute the complexities of other ingredients that will be directly used in Algorithm 1, which are listed in Table I.

During each iteration of the RCG method, the Euclidean gradient $\mathrm{grad}_{\mathcal{N}}f(\mathbf{P}^p)$ needs to be calculated once according to (74), where $\hat{\mathbf{E}}_{\ell k}^p$, $\check{\mathbf{R}}_{\ell n}^p$ and $\mathbf{R}_{\ell n}^p$ can be directly obtained from the $(p-1)$-th iteration. The computational complexity of $\mathrm{grad}_{\mathcal{N}}f(\mathbf{P}^p)$ is $O(M_tN_SN_R+M_t^2\log(N_t)+M_t^2N_S)$. Then, the update direction $\boldsymbol{\eta}_n^p$ can be obtained from (54), with which the step length can be searched in the inner iteration. Before entering the inner iteration for searching for the step length, $\bar{\mathbf{E}}_{k\ell}^p$, $\bar{\mathbf{F}}_{k\ell}^p$ and $\tilde{\mathbf{F}}_{k\ell}^p$ are precomputed to

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

TABLE II
DETAILED SIMULATION PARAMETERS

f_c	4.8 GHz	v	300 km/h
M_t	128	M_r	2
N_c	2048	N_g	144
N_v	120	Δf	60 KHz
N_b	4	N_s	14
$P_u, \forall u \in \mathcal{U}$	7	$Q_{p,u}, p \in \mathcal{S}_u^{\mathrm{NLoS}}, u \in \mathcal{U}$	10
Δ_r	$\frac{1}{2}$	Δ_t	$\frac{1}{2}$
F_r	1	F_t	2
F_{de}	2	F_{do}	1, 2, 4
$d_k, \forall k \in \mathcal{U}$	2	$w_k, \forall k \in \mathcal{U}$	1
Ricean K-factor	7 dB	BS-track distance	200 m
Height of each BS	25 m	Height of each UT	1.5 m

facilitate the computation of the objective function during the inner iteration. During the inner iteration, the objective function needs to be computed per inner iteration, whose input $\mathbf{R}_{kn}^{p,q}$ and $\check{\mathbf{R}}_{kn}^{p,q}$ can be directly obtained with (72), (73) and the precomputed matrices. In addition, the complexity of computing $\log\det(\cdot)$ in the objective function can be omitted as the dimension of the input is M_r, which is typically a small number, indicating that the objective function can be computed at minimal computational cost.

Therefore, the computational complexity of the RCG method is $O(N_t^2 + 2M_tN_SN_R + 2N_t\log(M_t)N_S + M_t^2\log(N_t) + M_t^2N_S)N_{\mathrm{out}}$. In the case that the predicted channel (21) is directly used for precoding without considering the prediction error, the computational complexity of implementing RCG on $\mathcal{M}$ is $O(2M_tN_SN_R)N_{\mathrm{out}}$. The method proposed in [25] also aims at maximizing the upper bound of the ergodic WSR based on the channel estimated from the pilot segments via minorize-maximization (MM) methodology, whose computational complexity is $O(M_t^2\log(N_t) + 2N_t\log(M_t)N_S + M_t^2N_S + M_t^3/2)N_{\mathrm{out}}$. Compared with this MM based method, our proposed RCG method avoids the inverse of large dimensional matrices and exhibits a lower complexity.

V. SIMULATION RESULTS

In this section, we provide the simulation results to validate the effectiveness of the channel model and the proposed RCG method for robust precoder design against channel aging and prediction error in the viaduct scenario. The channel estimated from the pilot segments (18) is termed as the pilot segment channel, and the channels predicted from (21) and (31) are referred to as the predicted channel and *a posterior* channel, respectively. The WSR performance is evaluated by applying the designed precoders to the real-time channel.

To establish a simulation environment that reproduces real-world characteristics of the HSR, we adopt the prevalent QuaDRiGa channel model [51] to generate a simulation scenario, where the "3GPP 37.885 Highway LOS" is considered to simulate the viaduct scenario. The HSR is programmed to travel eastward along the x-axis at a speed of v = 300 km/h with $U = 20$ UTs uniformly spaced at 10-meter intervals on the HSR. The initial position of the first UT is set at

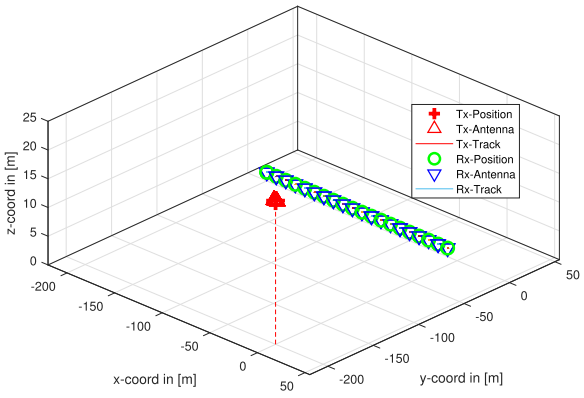


Fig. 3. The layout of the HSR QuaDriGa model.

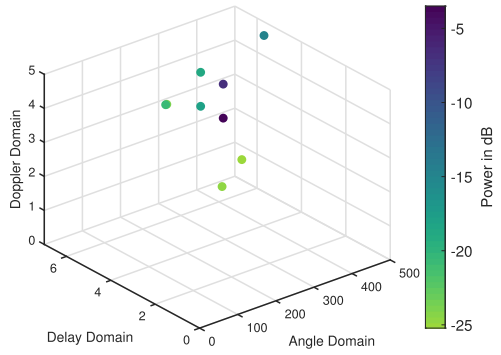


Fig. 4. The power density of the QB domain channel matrix.

coordinates (-200,0) and the vertical BS-track distance is set to 200 m with the BS positioned at (0,-200). The layout of this massive MIMO system for HSR communications is plotted in Fig. 3. The time duration of each subframe is set to 1 ms and the subcarrier spacing is set to 60 KHz according to [52]. In this case, each subframe consists of $N_b = 4$ time slots and each time slot is made up of $N_s = 14$ OFDM symbols. More detailed simulation parameters are available in Table II. As it is assumed that the CSI keeps constant within an OFDM symbol time duration and varies symbol by symbol, the HSR are tracked uniformly along the rail for a time duration of 2 ms to generate the channels for each OFDM symbol. The power of the channel generated from QuaDRiGa is normalized and the transmit SNR is defined as $\frac{P_t}{\sigma_z^2}$ with $\sigma_z^2 = 1$.

First of all, to demonstrate that the QB domain channel in the viaduct scenario does exhibit the sparse property, the power density of the QB domain channel matrix of one UT is plotted in Fig. 4, where the entries of negligible magnitude are set to zero for clarity. As can be clearly seen from Fig. 4, the energy of the QB domain channel matrix is predominantly concentrated on a small subset of elements, particularly within the angular domain, thereby affirming the sparse nature of the QB domain channel and validating the rationale behind our proposed channel model.

Then, we compare the normalized mean-squared error (NMSE) performance of the predicted channel (20) and the

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

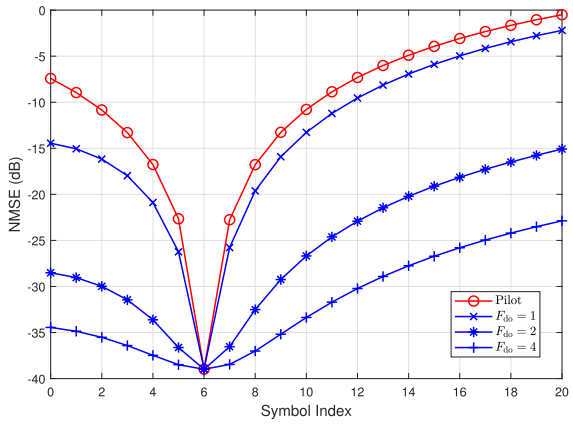


Fig. 5. The comparison of NMSE performance versus OFDM symbol index.

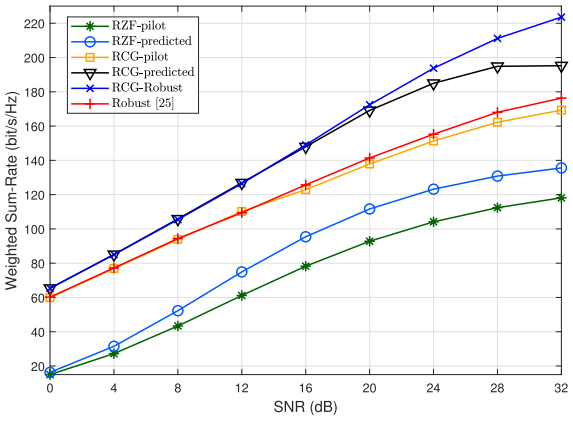


Fig. 7. The WSR performance of the RCG method at the fourteenth predicted OFDM symbol.

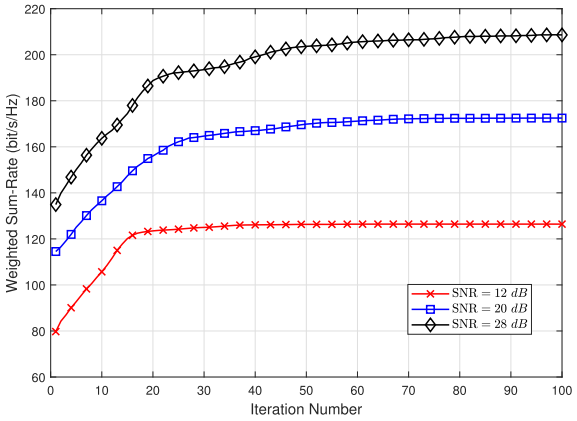


Fig. 6. The convergence behavior of the RCG method at different SNRs.

pilot segment channel (18) in Fig. 5, where the sixth OFDM index is the pilot symbol. By observing Fig. 5, we can see that directly applying the pilot segment channel to the data symbol has a much poorer NMSE performance compared with the predicted channel. In addition, the NMSE performance of the pilot segment channel significantly degrades as the temporal separation between the data symbol and the pilot symbol increases, thereby highlighting the necessity of introducing the channel prediction technique. For channel prediction, we further compare the NMSE performance of the predicted channel at different F_{do}. As shown in Fig. 5, the NMSE improves with the increase of F_{do}, which confirms the ability of higher F_{do} in the enhancement of channel prediction accuracy. However, increasing F_{do} also introduces increased computational complexity. Therefore, in subsequent simulations, we set $F_{\mathrm{do}} = 2$, as this value can guarantee good enough prediction performance. Although the predicted channel exhibits a significant performance gain compared with the channel without prediction, the NMSE performance still unavoidably declines as the delay between the pilot symbol and data symbols increases. This observation explains the motivation behind developing the *a posterior* channel model.

We then study the convergence behavior of the proposed RCG method for robust precoder design at the fourteenth OFDM symbol after the pilot symbol in Fig. 6. Note that the WSR performance in subsequent simulations is averaged over the valid sub-carriers. As we can see from Fig. 6, the RCG method has achieved 79% and 90% performance gains in the first 10 and 20 iterations, respectively, when SNR=20 dB. We also observe that RCG method needs about 20, 40 and 60 iterations to converge for SNR=12 dB, SNR=20 dB and SNR=28 dB, respectively.

To show the superiority of the proposed RCG method for robust precoding, we compare the WSR performance of the RCG method with those of other approaches at the fourteenth OFDM symbol after the pilot symbol, which is 0.25 ms away from the pilot symbol. For clarity, the precoders designed by the RCG method with the pilot segment channel (18), the predicted channel (21) and the *a posterior* channel (31) are termed as the RCG-pilot precoder, the RCG-predicted precoder and the RCG-robust precoder, respectively. Similarly, the precoders designed by the regularized zero-forcing (RZF) method with the pilot segment channel and the predicted channel are termed as the RZF-pilot precoder and RZF-predicted precoder, respectively. As shown in Fig. 7, the RCG-predicted precoder has a much better performance compared with that of the RCG-pilot precoder, showing the advantage of our predicted channel. Additionally, although the RZF-predicted precoder outperforms the RZF-pilot precoder, it still has a much performance loss compared with the precoders designed by the RCG method, showing the superiority of the matrix manifold optimization. We also observe that the RCG-robust precoder outperforms the RCG-predicted precoder, especially in the high SNR regime, demonstrating the effectiveness of the proposed *a posterior* channel model and the RCG method for robust precoding. In addition, the robust precoder designed from the RCG method exhibits significant performance gain compared with that of the robust precoder designed by the method proposed in [25], further validating the superiority of our proposed *a posterior* channel model and the RCG method.

It is noteworthy that we can predict 14 OFDM symbols under the simulation parameter configurations listed in Table II, corresponding to the channels within 0.25 ms from the pilot symbol. Indeed, the proposed *a posterior* channel model allows for channel prediction over extended time intervals, which can effectively save the pilot resources while

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

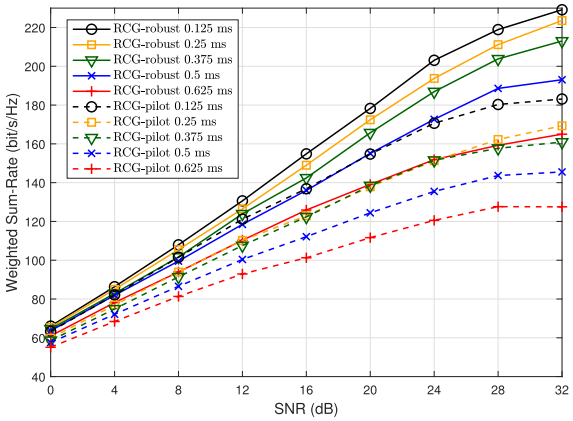


Fig. 8. The WSR performance of the RCG method with different predicted channel.

maintain the performance. To show the long-term predictive capability of our channel model, the WSR performances of the precoders designed with the channels predicted for different segments are shown in Fig. 8. In addition, the WSR performances of applying the precoder designed at the pilot symbol to data symbols are plotted for comparison. For brevity, the precoder designed by the RCG method with the *a posterior* channel 0.5 ms away from the pilot symbol is termed as RCG-robust 0.5 ms precoder. Similarly, applying the precoder designed for the pilot segment channel to the data symbol with a delay of 0.5 ms from the pilot symbol with the RCG method [38] is refereed to as the RCG-pilot 0.5 ms precoder. By observing Fig. 8, we can see that the WSR performance inevitably degrades with the extension of the prediction time. In spite of this, RCG-robust precoder still exhibits a satisfactory performance gain compared with that of the RCG-pilot precoder in all the cases. To be specific, when SNR=20 dB, the WSR performance of the RCG-robust 0.5 ms precoder has a 24% performance gain compared with RCG-pilot 0.5 ms precoder. In addition, it is noteworthy that the WSR performance of the RCG-robust 0.5 ms precoder still outperforms that of the RCG-pilot 0.125 ms precoder. These observations demonstrate that the WSR performance can be effectively preserved with decreased pilot overhead by virtue of our proposed *a posterior* channel model and the corresponding RCG method.

VI. Conclusion

In this paper, we investigated the robust precoder design for massive MIMO HSR communications with matrix manifold optimization. The QB based channel model was established for acquiring a more accurate CSI. To further improve the performance, the QB domain channels estimated from different virtual subframes were treated as a time series, and the *a posterior* QB based statistical channel model was established to reduce the prediction error, with which the channel for the forthcoming OFDM symbols can be better predicted. With the matrix manifold optimization, the ergodic WSR-maximization problem was formulated on the Riemannian submanifold formed by the precoders satisfying TPC, and the Riemannian ingredients were derived. With the proposed *a posterior* QB model and the derived Riemannian ingredients, the RCG method was proposed for solving the unconstrained optimization problem on the Riemannian submanifold. The expectation operators were avoided with the upper bound of the ergodic WSR and the QB domain statistical CSI, and the inverses of large dimensional matrices were avoided with the RCG method. The complexity analyses demonstrated the computational efficiency of the proposed RCG method. The simulation results verified the high accuracy of the predicted channel, the effectiveness of the *a posterior* channel model against channel aging and prediction inaccuracy, and the superiority of the RCG method for robust precoder design for the massive MIMO HSR system.

Appendix A
Proof for Theorem 1

With a little abuse of notation, $f(\mathbf{P}_{kn})$ is utilized to represent $f(\mathbf{P}_n)$ that only considers $\mathbf{P}_{kn}$ as the variable with $\mathbf{P}_{\ell n}, \ell \neq k$, fixed. The Euclidean gradient $\mathrm{grad} f(\mathbf{P}_{kn})$, which is also termed as the Riemannian gradient on $\mathcal{N}_k = \mathbb{C}^{M_t \times d_k}$, is defined as the unique element in $T_{\mathbf{P}_{kn}}\mathcal{N}_k$ that satisfies

$$g_{\mathbf{P}_{kn}}\left(\mathrm{grad} f(\mathbf{P}_{kn}), \boldsymbol{\xi}_{\mathbf{P}_{kn}}\right) = \mathrm{D} f(\mathbf{P}_{kn})\left[\boldsymbol{\xi}_{\mathbf{P}_{kn}}\right], \tag{60}$$

where $\mathrm{D} f(\mathbf{P}_{kn})\left[\boldsymbol{\xi}_{\mathbf{P}_{kn}}\right]$ is the directional derivative of $f(\mathbf{P}_{kn})$ along $\boldsymbol{\xi}_{\mathbf{P}_{kn}}$. For our considered objective function and the Euclidean space $\mathcal{N}_k$, $\mathrm{D} f(\mathbf{P}_{kn})\left[\boldsymbol{\xi}_{\mathbf{P}_{kn}}\right]$ reduces to the classical directional derivative and is given by

$$\begin{aligned}
&\mathrm{D} f(\mathbf{P}_{kn})\left[\boldsymbol{\xi}_{\mathbf{P}_{kn}}\right] \\
&= \lim_{t\to 0} \frac{f\left(\mathbf{P}_{kn} + t\boldsymbol{\xi}_{\mathbf{P}_{kn}}\right) - f(\mathbf{P}_{kn})}{t} \\
&= -w_i \mathrm{D}\mathcal{R}_{kn}^{\mathrm{ub}}\left[\boldsymbol{\xi}_{\mathbf{P}_{kn}}\right] - \sum_{\ell \neq k} w_\ell \mathrm{D}\mathcal{R}_{\ell n}^{\mathrm{ub}}\left[\boldsymbol{\xi}_{\mathbf{P}_{kn}}\right].
\end{aligned} \tag{61}$$

$\mathrm{D}\mathcal{R}_{kn}^{\mathrm{ub}}\left[\boldsymbol{\xi}_{\mathbf{P}_{kn}}\right]$ and $\mathrm{D}\mathcal{R}_{\ell n}^{\mathrm{ub}}\left[\boldsymbol{\xi}_{\mathbf{P}_{kn}}\right]$ are derived separately as

$$\begin{aligned}
&\mathrm{D}\mathcal{R}_{\ell n}^{\mathrm{ub}}\left[\boldsymbol{\xi}_{\mathbf{P}_{\ell n}}\right] \\
&= 2\Re\left\{\mathrm{tr}\left(\mathbb{E}_{\mathbf{H}_{\ell n}}\left\{\mathbf{H}_{\ell n}^H\left(\check{\mathbf{R}}_{\ell n}^{-1} - \mathbf{R}_{\ell n}^{-1}\right)\mathbf{H}_{\ell n}\right\}\mathbf{P}_{kn}\boldsymbol{\xi}_{\mathbf{P}_{kn}}^H\right)\right\}.
\end{aligned} \tag{62}$$

and

$$\begin{aligned}
&\mathrm{D}\mathcal{R}_{\ell n}^{\mathrm{ub}}\left[\boldsymbol{\xi}_{\mathbf{P}_{\ell n}}\right] \\
&= 2\Re\left\{\mathrm{tr}\left(\mathbb{E}_{\mathbf{H}_{\ell n}}\left\{\mathbf{H}_{\ell n}^H\left(\check{\mathbf{R}}_{\ell n}^{-1} - \mathbf{R}_{\ell n}^{-1}\right)\mathbf{H}_{\ell n}\right\}\mathbf{P}_{kn}\boldsymbol{\xi}_{\mathbf{P}_{kn}}^H\right)\right\}.
\end{aligned} \tag{63}$$

With (60), (62) and (63), we can obtain (46).

The tangent space $T_{\mathbf{P}_n}\mathcal{N}$ can be decomposed into

$$T_{\mathbf{P}_n}\mathcal{N} = T_{\mathbf{P}_n}\mathcal{M} \oplus N_{\mathbf{P}_n}\mathcal{M}. \tag{64}$$

For the Riemannian submanifold $\mathcal{M}$, $T_{\mathbf{P}_n}\mathcal{M}$ is the kernel of the differential of $F(\mathbf{P}_n)$ and can be expressed as

$$\begin{aligned}
T_{\mathbf{P}_n}\mathcal{M} &= \ker\left(\mathrm{D}F(\mathbf{P}_n)\right) \\
&= \left\{\boldsymbol{\xi}_{\mathbf{P}_n} \in T_{\mathbf{P}_n}\mathcal{N} \mid g_{\mathbf{P}_n}\left(\mathbf{P}_n, \boldsymbol{\xi}_{\mathbf{P}_n}\right) = 0\right\}.
\end{aligned} \tag{65}$$

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

$N_{\mathbf{P}_n}\mathcal{M}$ is the orthogonal complement of $T_{\mathbf{P}_n}\mathcal{M}$ and given by

$$N_{\mathbf{P}_n}\mathcal{M}=\left\{\boldsymbol{\xi}_{\mathbf{P}_n}\in T_{\mathbf{P}_n}\mathcal{N}\,|\,g_{\mathbf{P}_n}\left(\boldsymbol{\xi}_{\mathbf{P}_n},\boldsymbol{\zeta}_{\mathbf{P}_n}\right)=0,\boldsymbol{\zeta}_{\mathbf{P}_n}\in T_{\mathbf{P}_n}\mathcal{M}\right\} =\lambda\mathbf{P}_n, \tag{66}$$

where $\lambda\in\mathbb{R}$. Then, we have $\left(\mathrm{grad}f\left(\mathbf{P}_n\right)-\lambda\mathbf{P}_n\right)\in T_{\mathbf{P}_n}\mathcal{M}$, which must satisfy (65). Then, we can derive that

$$\lambda=\frac{1}{P}\Re\left\{\mathrm{tr}\left(\mathbf{P}_n^H\mathrm{grad}_{\mathcal{N}}f\left(\mathbf{P}_n\right)\right)\right\}. \tag{67}$$

Appendix B
Detailed Derivations for Efficient Algorithm Implementation

With (51), we have

$$\mathbf{P}_{\ell n}^{p,q}\left(\mathbf{P}_{\ell n}^{p,q}\right)^H=\left(\rho^{p,q}\right)^2\Big(\mathbf{P}_{\ell n}^{p}\left(\mathbf{P}_{\ell n}^{p}\right)^H+\alpha^{p,q}\mathbf{P}_{\ell n}^{p}(\boldsymbol{\eta}_{\ell n}^{p})^H +\alpha^{p,q}\boldsymbol{\eta}_{\ell n}^{p}\left(\mathbf{P}_{\ell n}^{p}\right)^H+\left(\alpha^{p,q}\right)^2\boldsymbol{\eta}_{\ell n}^{p}(\boldsymbol{\eta}_{\ell n}^{p})^H\Big). \tag{68}$$

Therefore, $\boldsymbol{\Phi}_{kn}\left(\mathbf{P}_{\ell n}^{p,q}\left(\mathbf{P}_{\ell n}^{p,q}\right)^H\right)$ can be obtained from

$$\boldsymbol{\Phi}_{kn}\left(\mathbf{P}_{\ell n}^{p,q}\left(\left(\mathbf{P}_{\ell n}^{p,q}\right)^H\right)\right)=\left(\rho^{p,q}\right)^2\Big(\hat{\mathbf{F}}_{k\ell}^{p}+\alpha^{p,q}\bar{\mathbf{F}}_{k\ell}^{p} +\alpha^{p,q}\left(\bar{\mathbf{F}}_{k\ell}^{p}\right)^H+\left(\alpha^{p,q}\right)^2\tilde{\mathbf{F}}_{k\ell}^{p}\Big), \tag{69}$$

where

$$\hat{\mathbf{F}}_{k\ell}^{p}=\boldsymbol{\Phi}_{kn}\left(\mathbf{P}_{\ell n}^{p}\left(\mathbf{P}_{\ell n}^{p}\right)^H\right), \tag{70a}$$

$$\bar{\mathbf{F}}_{k\ell}^{p}=\boldsymbol{\Phi}_{kn}\left(\mathbf{P}_{\ell n}^{p}\left(\boldsymbol{\eta}_{\ell n}^{p}\right)^H\right), \tag{70b}$$

$$\tilde{\mathbf{F}}_{k\ell}^{p}=\boldsymbol{\Phi}_{kn}\left(\boldsymbol{\eta}_{\ell n}^{p}\left(\boldsymbol{\eta}_{\ell n}^{p}\right)^H\right). \tag{70c}$$

Note that $\hat{\mathbf{F}}_{k\ell}^{p}$, $\bar{\mathbf{F}}_{k\ell}^{p}$ and $\tilde{\mathbf{F}}_{k\ell}^{p}$ are independent of the index of inner iterations, indicating that (69) can be effectively computed during the inner iteration with the precomputed elements. Moreover, let $\hat{\mathbf{E}}_{k\ell}^{p,q}=\hat{\mathbf{H}}_{kn}\mathbf{P}_{\ell n}^{p,q}\in\mathbb{C}^{M_r\times d_k}$. With (51), $\hat{\mathbf{E}}_{k\ell}^{p,q}$ can be computed by

$$\hat{\mathbf{E}}_{k\ell}^{p,q}=\rho^{p,q}\hat{\mathbf{H}}_{kn}\left(\mathbf{P}_{\ell n}^{p}+\alpha^{p,q}\boldsymbol{\eta}_{\ell n}^{p}\right), \tag{71}$$

indicating that $\hat{\mathbf{E}}_{k\ell}^{p,q}$ can be effectively computed during the inner iteration with the precomputed $\hat{\mathbf{E}}_{k\ell}^{p}=\hat{\mathbf{H}}_{kn}\mathbf{P}_{\ell n}^{p}$ and $\bar{\mathbf{E}}_{k\ell}^{p}=\hat{\mathbf{H}}_{kn}\boldsymbol{\eta}_{\ell n}^{p}$, which are all irrelevant to the index of inner iterations. With (69) and (71), $\mathbf{R}_{kn}^{p,q}$ (34) and $\check{\mathbf{R}}_{kn}^{p,q}$ (44) can be rewritten as

$$\mathbf{R}_{kn}^{p,q}=\sum_{\ell\neq k}\hat{\mathbf{E}}_{k\ell}^{p,q}\left(\hat{\mathbf{E}}_{k\ell}^{p,q}\right)^H+\sum_{\ell\neq k}\boldsymbol{\Phi}_{kn}\Big(\mathbf{P}_{\ell n}^{p,q}\left(\mathbf{P}_{\ell n}^{p,q}\right)^H\Big)+\sigma_z^2\mathbf{I}_{M_r} \tag{72}$$

and

$$\check{\mathbf{R}}_{kn}^{p,q}=\sum_{\ell\in\mathcal{U}}\hat{\mathbf{E}}_{k\ell}^{p,q}\left(\hat{\mathbf{E}}_{k\ell}^{p,q}\right)^H+\sum_{\ell\in\mathcal{U}}\boldsymbol{\Phi}_{kn}\Big(\mathbf{P}_{\ell n}^{p,q}\left(\mathbf{P}_{\ell n}^{p,q}\right)^H\Big)+\sigma_z^2\mathbf{I}_{M_r}, \tag{73}$$

respectively. With the precomputed $\hat{\mathbf{F}}_{k\ell}^{p}$, $\bar{\mathbf{F}}_{k\ell}^{p}$, $\tilde{\mathbf{F}}_{k\ell}^{p}$, $\hat{\mathbf{E}}_{k\ell}^{p}$ and $\bar{\mathbf{E}}_{k\ell}^{p}$, $\mathbf{R}_{kn}^{p,q}$ (72) and $\check{\mathbf{R}}_{kn}^{p,q}$ (73) can be derived through a weighted sum of known matrices, showing that the objective function can be efficiently computed during the inner iteration.

After reaching a good point in the (p,q)-th iteration, $f\left(\mathbf{P}_n^{p+1}\right)=f\left(\mathbf{P}_n^{p,q}\right)$, $\hat{\mathbf{E}}_{k\ell}^{p+1}=\hat{\mathbf{E}}_{k\ell}^{p,q}$, $\hat{\mathbf{F}}_{k\ell}^{p+1}=\boldsymbol{\Phi}_{kn}\left(\mathbf{P}_{\ell n}^{p,q}\left(\left(\mathbf{P}_{\ell n}^{p,q}\right)^H\right)\right)$, $\mathbf{R}_{kn}^{p+1}=\mathbf{R}_{kn}^{p,q}$ and $\check{\mathbf{R}}_{kn}^{p+1}=\check{\mathbf{R}}_{kn}^{p,q}$ can be seamlessly utilized as the input of the next iteration. Then, the Euclidean gradient (46) in the $(p+1)$-th iteration can be further written as

$$\begin{aligned}&\mathrm{grad}_{\mathcal{N}}f\left(\mathbf{P}_{kn}^{p+1}\right)\\&=-2\sum_{\ell\in\mathcal{U}}\left(\hat{\mathbf{H}}_{\ell n}^H\left(\check{\mathbf{R}}_{\ell n}^{p+1}\right)^{-1}\hat{\mathbf{E}}_{\ell k}^{p+1}+\boldsymbol{\Psi}_{\ell n}\left(\left(\check{\mathbf{R}}_{\ell n}^{p+1}\right)^{-1}\right)\mathbf{P}_{kn}^{p+1}\right)\\&\quad+2\sum_{\ell\neq k}\left(\hat{\mathbf{H}}_{\ell n}^H\left(\mathbf{R}_{\ell n}^{p+1}\right)^{-1}\hat{\mathbf{E}}_{\ell k}^{p+1}+\boldsymbol{\Psi}_{\ell n}\left(\left(\mathbf{R}_{\ell n}^{p+1}\right)^{-1}\right)\mathbf{P}_{kn}^{p+1}\right),\end{aligned} \tag{74}$$

where $\hat{\mathbf{E}}_{\ell k}^{p+1}$, $\check{\mathbf{R}}_{\ell n}^{p+1}$ and $\mathbf{R}_{\ell n}^{p+1}$ all can be retrieved from the outputs of the previous iteration without incurring additional complexity.

References

[1] B. Ai et al., "Future railway services-oriented mobile communications network," *IEEE Commun. Mag.*, vol. 53, no. 10, pp. 78–85, Oct. 2015.

[2] C.-X. Wang, A. Ghazal, B. Ai, Y. Liu, and P. Fan, "Channel measurements and models for high-speed train communication systems: A survey," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 2, pp. 974–987, 2nd Quart., 2016.

[3] Z. Zou, M. Careem, A. Dutta, and N. Thawdar, "Joint spatio-temporal precoding for practical non-stationary wireless channels," *IEEE Trans. Commun.*, vol. 71, no. 4, pp. 2396–2409, Apr. 2023.

[4] M. Gao et al., "IRS-assisted high-speed train communications: Performance analysis and optimal configuration," *IEEE Internet Things J.*, vol. 10, no. 21, pp. 18980–18992, Nov. 2023.

[5] Y. Ma et al., "Robust train-to-train transmission scheduling in mmWave band for high speed train communication systems," *IEEE Trans. Veh. Technol.*, vol. 72, no. 12, pp. 16148–16162, Dec. 2023.

[6] Y. Qiao et al., "Intelligent beam management based on deep reinforcement learning in high-speed railway scenarios," *IEEE Trans. Veh. Technol.*, vol. 73, no. 3, pp. 3917–3931, Mar. 2024.

[7] A. Ghazal, C.-X. Wang, B. Ai, D. Yuan, and H. Haas, "A nonstationary wideband MIMO channel model for high-mobility intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 2, pp. 885–897, Apr. 2015.

[8] C.-X. Wang et al., "On the road to 6G: Visions, requirements, key technologies and testbeds," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 905–974, 2nd Quart., 2023.

[9] Q. Wu et al., "A comprehensive overview on 5G-and-beyond networks with UAVs: From communications to sensing and intelligence," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 10, pp. 2912–2945, Oct. 2021.

[10] R. M. Dreifuerst and R. W. Heath, "Massive MIMO in 5G: How beamforming, codebooks, and feedback enable larger arrays," *IEEE Commun. Mag.*, vol. 61, no. 12, pp. 18–23, Dec. 2023.

[11] M. A. Albreem, M. Juntti, and S. Shahabuddin, "Massive MIMO detection techniques: A survey," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3109–3132, 4th Quart., 2019.

[12] X. L. Yu, X. Q. Gao, A.-A. Lu, J. Zhang, W. Hebing, and G. Y. Li, "Robust precoding for HF skywave massive MIMO," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6691–6705, Oct. 2023.

[13] L. You, J. Xu, G. C. Alexandropoulos, J. Wang, W. Wang, and X. Gao, "Energy efficiency maximization of massive MIMO communications with dynamic metasurface antennas," *IEEE Trans. Wireless Commun.*, vol. 22, no. 1, pp. 393–407, Jan. 2023.

[14] M. Gao et al., "Efficient hybrid beamforming with anti-blockage design for high-speed railway communications," *IEEE Trans. Veh. Technol.*, vol. 69, no. 9, pp. 9643–9655, Sep. 2020.

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.

[15] Y. Liu et al., "Adaptive non-uniform hybrid beamforming for mmWave train-to-ground communications in high-speed railway scenarios," *IEEE Trans. Veh. Technol.*, vol. 72, no. 7, pp. 9085–9098, Jul. 2023.

[16] Q. Ding, J. Luo, H. Shi, and T. Fu, "RIS-aided hybrid precoder design with anti-blocking scheme in high-speed railway," *IEEE Trans. Veh. Technol.*, vol. 72, no. 3, pp. 3335–3344, Mar. 2023.

[17] A.-A. Lu, Y. Chen, and X. Gao, "2D beam domain statistical CSI estimation for massive MIMO uplink," *IEEE Trans. Wireless Commun.*, vol. 23, no. 1, pp. 749–761, Jan. 2024.

[18] J. Shi, A.-A. Lu, W. Zhong, X. Gao, and G. Y. Li, "Robust WMMSE precoder with deep learning design for massive MIMO," *IEEE Trans. Commun.*, vol. 71, no. 7, pp. 3963–3976, Jul. 2023.

[19] S. Han et al., "Achieving high spectrum efficiency on high speed train for 5G new radio and beyond," *IEEE Wireless Commun.*, vol. 26, no. 5, pp. 62–69, Oct. 2019.

[20] A. O. Laiyemo, H. Pennanen, P. Pirinen, and M. Latva-aho, "Transmission strategies for throughput maximization in high-speed-train communications: From theoretical study to practical algorithms," *IEEE Trans. Veh. Technol.*, vol. 66, no. 4, pp. 2997–3011, Apr. 2017.

[21] R. Sun, C. Tao, L. Liu, Z. Tan, L. Zhang, and T. Zhou, "Nonisotropic scattering characteristic in an alternant tree-blocked viaduct scenario on high-speed railway at 2.35 GHz," *Int. J. Antennas Propag.*, vol. 2014, no. 1, 2014, Art. no. 642894.

[22] Y. Liu, C. Wang, and J. Huang, "Recent developments and future challenges in channel measurements and models for 5G and beyond high-speed train communication systems," *IEEE Commun. Mag.*, vol. 57, no. 9, pp. 50–56, Sep. 2019.

[23] J. Zhang, H. Du, P. Zhang, J. Cheng, and L. Yang, "Performance analysis of 5G mobile relay systems for high-speed trains," *IEEE J. Sel. Area. Commun.*, vol. 38, no. 12, pp. 2760–2772, Dec. 2020.

[24] A.-A. Lu, X. Gao, W. Zhong, C. Xiao, and X. Meng, "Robust transmission for massive MIMO downlink with imperfect CSI," *IEEE Trans. Commun.*, vol. 67, no. 8, pp. 5362–5376, Aug. 2019.

[25] A. Lu, X. Gao, and C. Xiao, "Robust linear precoder design for 3D massive MIMO downlink with a posteriori channel model," *IEEE Trans. Veh. Technol.*, vol. 71, no. 7, pp. 7274–7286, Jul. 2022.

[26] Y. Liu, S. Zhang, F. Gao, J. Ma, and X. Wang, "Uplink-aided high mobility downlink channel estimation over massive MIMO-OTFS system," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 9, pp. 1994–2009, Sep. 2020.

[27] M. Li et al., "A new path division multiple access for the massive MIMO-OTFS networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 4, pp. 903–918, Apr. 2021.

[28] C. Wu, X. Yi, Y. Zhu, W. Wang, L. You, and X. Gao, "Channel prediction in high-mobility massive MIMO: From spatio-temporal autoregression to deep learning," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 7, pp. 1915–1930, Jul. 2021.

[29] A. Careem, M. Ahamed, and A. Dutta, "Real-time prediction of non-stationary wireless channels," *IEEE Trans. Wireless Commun.*, vol. 19, no. 12, pp. 7836–7850, Dec. 2020.

[30] D. Shi, L. Song, W. Zhou, X. Q. Gao, C.-X. Wang, and G. Y. Li, "Channel acquisition for HF skywave massive MIMO-OFDM communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 4074–4089, Jun. 2023.

[31] Z. Wang, J. Zhang, B. Ai, C. Yuen, and M. Debbah, "Uplink performance of cell-free massive MIMO with multi-antenna users over jointly-correlated Rayleigh fading channels," *IEEE Trans. Wireless Commun.*, vol. 21, no. 9, pp. 7391–7406, Sep. 2022.

[32] C. Wang, A. Lu, X. Gao, and Z. Ding, "Robust precoding for 3D massive MIMO configuration with matrix manifold optimization," *IEEE Trans. Wireless Commun.*, vol. 21, no. 5, pp. 3423–3437, May 2022.

[33] T. Li et al., "Joint device detection, channel estimation, and data decoding with collision resolution for MIMO massive unsourced random access," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 5, pp. 1535–1555, May 2022.

[34] E. Shtaiwi, H. Zhang, A. Abdelhadi, A. Lee Swindlehurst, Z. Han, and H. Vincent Poor, "Sum-rate maximization for RIS-assisted integrated sensing and communication systems with manifold optimization," *IEEE Trans. Commun.*, vol. 71, no. 8, pp. 4909–4923, Aug. 2023.

[35] W. Yin, Z. Ma, and Q. Liu, "Discriminative subspace learning via optimization on Riemannian manifold," *Pattern Recognit.*, vol. 139, Jul. 2023, Art. no. 109450.

[36] R. Sun, L. You, A.-A. Lu, C. Sun, X. Gao, and X.-G. Xia, "Precoder design for user-centric network massive MIMO with matrix manifold optimization," 2024, *arXiv:2404.07425*.

[37] B. Shustin and H. Avron, "Riemannian optimization with a preconditioning scheme on the generalized Stiefel manifold," *J. Comput. Appl. Math.*, vol. 423, May 2023, Art. no. 114953.

[38] R. Sun, C. Wang, A.-A. Lu, X. Gao, and X.-G. Xia, "Precoder design for massive MIMO downlink with matrix manifold optimization," *IEEE Trans. Signal Process.*, vol. 72, pp. 1065–1080, 2024.

[39] Z. Wang et al., "Extremely large-scale MIMO: Fundamentals, challenges, solutions, and future directions," *IEEE Wireless Commun.*, vol. 31, no. 3, pp. 117–124, Jun. 2023.

[40] B. Ai et al., "Challenges toward wireless communications for high-speed railway," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 5, pp. 2143–2158, Oct. 2014.

[41] *Study on Performance Enhancements for High Speed Scenario in LTE*, document 36.878, 3rd Gener. Partnership Project (3GPP), 2015.

[42] R. Alieiev, T. Hehn, A. Kwoczek, and T. Kürner, "Predictive communication and its application to vehicular environments: Doppler-shift compensation," *IEEE Trans. Veh. Technol.*, vol. 67, no. 8, pp. 7380–7393, Aug. 2018.

[43] M. Gümüş and T. M. Duman, "Channel estimation and symbol demodulation for OFDM systems over rapidly varying multipath channels with hybrid deep neural networks," *IEEE Trans. Wireless Commun.*, vol. 22, no. 12, pp. 9361–9373, Dec. 2023.

[44] J. Pearl, *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. San Francisco, CA, USA: Morgan Kaufmann Publishers Inc., 1988.

[45] J. Ma and L. Ping, "Orthogonal AMP," *IEEE Access*, vol. 5, pp. 2020–2033, 2017.

[46] J. Yang, A. Lu, Y. Chen, X. Gao, X. Xia, and D. T. M. Slock, "Channel estimation for massive MIMO: An information geometry approach," *IEEE Trans. Signal Process.*, vol. 70, pp. 4820–4834, 2022.

[47] Y. Liu, Y. Wang, J. Wang, L. You, W. Wang, and X. Gao, "Robust downlink precoding for LEO satellite systems with per-antenna power constraints," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10694–10711, Oct. 2022.

[48] P.-A. Absil, R. Mahony, and R. Sepulchre, *Optimization Algorithms on Matrix Manifolds*. Princeton, NJ, USA: Princeton Univ. Press, 2009.

[49] H. Sato, "Riemannian conjugate gradient methods: General framework and specific algorithms with convergence analyses," *SIAM J. Optim.*, vol. 32, no. 4, pp. 2690–2717, 2022.

[50] N. Boumal, *An Introduction to Optimization on Smooth Manifolds*. Cambridge, U.K.: Cambridge Univ. Press, 2023.

[51] S. Jaeckel, L. Raschkowski, K. Borner, and L. Thiele, "QuaDRiGa: A 3-D multi-cell channel model with time evolution for enabling virtual field trials," *IEEE Trans. Antennas Propag.*, vol. 62, no. 6, pp. 3242–3256, Jun. 2014.

[52] *5G; NR; Physical Channels and Modulation*, document 36.211, 3rd Gener. Partnership Project (3GPP), 2023.



Rui Sun (Graduate Student Member, IEEE) received the B.E. degree in communication engineering from Xidian University, Xi'an, China, in 2021. He is currently pursuing the Ph.D. degree with the National Mobile Communications Research Laboratory, Southeast University, Nanjing, China. His research interests include manifold optimization, signal processing for wireless communications, and massive MIMO wireless communications.



Chen Sun (Member, IEEE) received the Ph.D. degree in electrical engineering from Southeast University, Nanjing, China, in 2018.

Currently, he is an Associate Professor with the National Mobile Communications Research Laboratory, Southeast University. From September 2015 to August 2016, he was a Visiting Student with the Department of Electrical and Computer Engineering, University of California at Davis, USA. His research interests include communications and information theory, with an emphasis on massive MIMO communications and optical wireless communications.

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.



Ding Shi (Member, IEEE) received the B.E. and Ph.D. degrees in information science and engineering from Southeast University, Nanjing, China, in 2018 and 2024, respectively. From November 2022 to November 2023, he conducted visiting research with the Division of Information Science and Engineering, KTH Royal Institute of Technology, Stockholm, Sweden. He is currently a Post-Doctoral Researcher with the Purple Mountain Laboratories, Nanjing, China. His research interests include massive MIMO communications, HF communications, and signal processing for wireless communications.



An-An Lu (Member, IEEE) received the B.E., M.E., and Ph.D. degrees in electronic engineering from Southeast University, Nanjing, China, in 2006, 2012, and 2017, respectively.

He jointed the National Mobile Communications Research Laboratory, Southeast University, in June 2017, where he is currently an Associate Professor. From November 2014 to February 2016, he visited Missouri University of Science and Technology, Rolla, MO, USA. From 2006 to 2008, he was with the Research Department, Hejian Technology Company Ltd., Suzhou, China. His current research interests include information geometry and wireless communications.



Xiqi Gao (Fellow, IEEE) received the Ph.D. degree in electrical engineering from Southeast University, Nanjing, China, in 1997.

He joined the Department of Radio Engineering, Southeast University, in April 1992. Since May 2001, he has been a Professor of information systems and communications. From September 1999 to August 2000, he was a Visiting Scholar with Massachusetts Institute of Technology, Cambridge, and Boston University, Boston, MA, USA. From August 2007 to July 2008, he visited Darmstadt University of Technology, Darmstadt, Germany, as a Humboldt scholar. His current research interests include broadband multicarrier communications, massive MIMO wireless communications, satellite communications, optical wireless communications, information theory, and signal processing for wireless communications.

Dr. Gao received the Science and Technology Awards of the State Education Ministry of China in 1998, 2006, and 2009; the National Technological Invention Award of China in 2011; the Science and Technology Award of Jiangsu Province of China in 2014; and the 2011 IEEE Communications Society Stephen O. Rice Prize Paper Award in the field of communications theory. From 2007 to 2012, he served as an Editor for IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS. From 2009 to 2013, he served as an Associate Editor for IEEE TRANSACTIONS ON SIGNAL PROCESSING. From 2015 to 2017, he served as an Editor for IEEE TRANSACTIONS ON COMMUNICATIONS.



Xiang-Gen Xia (Fellow, IEEE) received the B.S. degree in mathematics from Nanjing Normal University, Nanjing, China, in 1983, the M.S. degree in mathematics from Nankai University, Tianjin, China, in 1986, and the Ph.D. degree in electrical engineering from the University of Southern California at Los Angeles, in 1992.

He was a Senior/Research Staff Member with the Hughes Research Laboratories, Malibu, CA, USA, from 1995 to 1996. In September 1996, he joined the Department of Electrical and Computer Engineering, University of Delaware, Newark, DE, USA, where he is the Charles Black Evans Professor. He is the author of the book *Modulated Coding for Intersymbol Interference Channels* (New York, Marcel Dekker, 2000). His current research interests include space-time coding, MIMO and OFDM systems, digital signal processing, and SAR and ISAR imaging.

Dr. Xia received the National Science Foundation (NSF) Faculty Early Career Development (CAREER) Program Award in 1997, the Office of Naval Research (ONR) Young Investigator Award in 1998, and the Outstanding Overseas Young Investigator Award from the National Nature Science Foundation of China in 2001. He also received the 2019 Information Theory Outstanding Overseas Chinese Scientist Award from the Information Theory Society of Chinese Institute of Electronics. He was the Technical Program Chair of the Signal Processing Symposium, Globecom 2007, Washington, DC, USA; and the General Co-Chair of ICASSP 2005, Philadelphia. He has served as an Associate Editor for several international journals, including IEEE TRANSACTIONS ON SIGNAL PROCESSING, IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, IEEE TRANSACTIONS ON MOBILE COMPUTING, and IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY.

Authorized licensed use limited to: Southeast University. Downloaded on June 18,2025 at 02:42:44 UTC from IEEE Xplore. Restrictions apply.